

Dyadic Scaling Rigidity and Log-Periodic Growth of 3AP-Free Permutations

Part II: Non-holonomicity of the counts

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Abstract

Let $\theta(n)$ count permutations of $[n]$ having no three-term arithmetic progression as a subsequence. A single exponential growth constant does not exist, by a theorem of Boon Suan Ho. We determine a more informative asymptotic object: a unique continuous function P , periodic in the binary logarithm, with

$$\log \theta(n) = nP(\log_2 n) - Q(n), \quad \log 2 \leq Q(n) \leq \log 21.$$

The construction is an application of established divide-and-conquer theory; an elementary proof with explicit constants is supplied. The central further result is an exact scale-rigidity theorem. Published enumerative data certify that P has fundamental period exactly one. Consequently, the positive real scales c for which $\log \theta(\lfloor cn \rfloor) - c \log \theta(n) = o(n)$ are precisely the integral powers of two. At every other scale, exponential gain and exponential loss both occur on sets of positive lower natural density. For the concrete split $3n = 2n + n$, integer certificates give a factor $(49/50)^n$ loss when $n = 128 \cdot 2^t$, and a factor $(21/20)^n$ gain when $n = 172 \cdot 2^t$. We also describe the entire interval of subsequential growth rates, all cutoff-dependent empirical limit laws, logarithmic averaging, and polynomial and geometric sampling. A finite-data enclosure algorithm, reproducible arithmetic certificates, an independent small-count verification, and a research program complete the article.

Research status. Unrefereed mathematical development, not a Lean-verified result. The general periodic-profile method and the failure of a single growth constant are credited to prior work. The scale-rigidity and statistical consequences are derived here; their priority in the entire literature has not been certified. Numerical certificates use published large counts, independently recomputed in this package only through $n = 32$.

Collection note [write]. This article is the report `a003407-dyadic-scaling-rigidity` of PROVEIT's research-report collection, written from one manuscript (batch 40, manuscript 06; see Section 14). None of its own theorems is formalized. Two inputs that the manuscript takes from the literature, Sharma's upper splitting bound and the parity lower recurrence, are already proved in Lean in PROVEIT; Section 11 names the declarations and says what they do and do not cover. Placement of this report near a Lean development gives it no formal status. Text added when the report was written is marked [write]. [Added 29 September 2026, batch 43: that manuscript is now Part I. Part II (Sections 15–27, Appendices B–C), from a second manuscript, proves that $\theta(n)$ and every arithmetic section $\theta(qn + r)$ are not P-recursive and that neither generating function is D-finite; it is not formalized either (Subsection 11.4).]

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1 The question and the contribution

1.1 What should replace a nonexistent growth constant?

Many counting problems are organized around an asymptotic expression of the form $C\rho^n n^\alpha$. For three-term-progression-free permutations, even the coarsest candidate $\rho = \lim \theta(n)^{1/n}$ fails to exist. Merely recording this failure leaves several natural structural questions unanswered. Does the sequence exhibit arbitrary exponential-scale oscillation, or a coherent phase law? Which changes of scale preserve its growth? Can differently sized blocks be combined at only subexponential cost? Does averaging eliminate the oscillation?

This article answers these questions in a unified way. The right object is a continuous growth profile on the circle $\mathbb{R}/\mathbb{Z}$, sampled at the phase $\log_2 n$. Its period is not just some divisor of one: a short finite certificate proves that its fundamental period is exactly one. This numerical shape information has a qualitative consequence for *every real scale*, not merely the finitely many sizes in the table.

Here is the central classification. In this display, $c > 0$ is fixed and the integer $\lfloor cn \rfloor$ is positive for all sufficiently large n :

$$\log \theta(\lfloor cn \rfloor) - c \log \theta(n) = o(n) \iff c \in \{2^j : j \in \mathbb{Z}\}. \quad (1.1)$$

Outside these scales, the left side has positive and negative excursions of order n , each occurring on a set of positive lower natural density. Thus the obstruction is substantially stronger than failure of a ratio limit.

1.2 Relationship to existing work

Davis, Entringer, Graham, and Simmons established the parity-splitting construction for these permutations [1]. Sharma proved the uniform upper splitting estimate used below [2, Theorem 2.8]. Correll and Randy W. Ho developed dynamic programming and extended the count table [4]; the OEIS table now supplies values through 200, with later terms attributed to Alois P. Heinz [7].

Boon Suan Ho proved in 2026 that $\theta(n)^{1/n}$ does not converge, by separating limits along two dyadic subsequences [5]. That theorem is *not* claimed as a result of this article. Separately, Hwang, Janson, and Tsai give a general theory of continuous periodic fluctuations for floor/ceiling divide-and-conquer recurrences [6, Theorem 2.10]. Our profile construction is the equal-weight, bounded-toll specialization of that theory. We give the proof because the explicit bounds and interpolation are needed for the subsequent certificates, not because the underlying analytic principle is new.

The additional contribution developed here consists of the exact period certificate, the resulting classification (1.1), exponential obstructions to unequal-size gluing, and the complete sampling and empirical laws for this particular counting sequence. These statements were not found in the sources inspected for this article. This is a targeted literature assessment, not a claim of exhaustive priority verification or of resolution of a separately named published conjecture.

1.3 Connection with the repository

The selected topic belongs directly to the Ramsey-theoretic part of PROVEIT [8]. In particular, the repository contains the directories

Combinatorics/Ramsey/Lean/DavisEntringerGrahamSimmons1977/

Combinatorics/Ramsey/Lean/LeSaulnierVijay2011/.

The latter statement catalogue, based on [3], uses $M(n)$ for the count denoted here by $\theta(n)$, records the even and odd parity recurrences, and includes the exponential bounds from the literature. A proposition-valued declaration in a statement catalogue is not by itself a proved theorem. We therefore take the published combinatorial estimates as the mathematical input, rather than inferring proof status from a declaration name.

The inspected repository revision was

74f7f5bdba1aa728b340509701a0fa4e45e8dbf3.

No repository files were changed, and no full Lean build or axiom audit was performed. The article extends the research direction, not the repository’s machine-checked theorem inventory.

Remark 1.1 (Correction [write]). The caution above is sound as method, but its conclusion is stale. At the inspected revision, and unchanged at the time of writing, the declarations in question are not bare statements. The two parity recurrences are proved as `LeanProofs.LeSaulnierVijay2011.M_even_recurrence_holds` and `M_odd_recurrence_holds`, which are derived from the Davis–Entringer–Graham–Simmons declarations `even_count_recurrence_holds` and `odd_count_recurrence_holds`. Sharma’s Theorem 2.8, which this article imports, is proved as `LeanProofs.Sharma2012.theorem_2_8_holds`. The manuscript read only the statement catalogues, so it did not see these proofs. Section 11 gives the exact names, files and scope. The mathematics below is unaffected: both inputs are true and are now also machine-checked in PROVEIT. The article’s own results remain unformalized.

1.4 Dependency and verification boundary

The one substantial theorem imported without reproducing its complete combinatorial proof is Sharma’s upper recurrence. All subsequent analytic and structural deductions are proved below. The exact-period and explicit-gluing certificates also use the published count table. The supplied program verifies the required integer inequalities, but that does not independently prove the large counts themselves. An independent subset-state enumeration was executed for every $0 \leq n \leq 32$, and direct permutation enumeration for $0 \leq n \leq 8$. These are separate checks with separate evidential roles.

[write] Sharma’s recurrence is imported by this article, but it is not an unchecked input in PROVEIT: its Lean proof is named in Section 11. The published count table beyond $n = 20$ has no counterpart in PROVEIT; the Lean development certifies $\theta(1), \dots, \theta(20)$ only, whereas the numerical enclosures of Section 4 and the certificates of Sections 5–6 use published counts with $100 \leq n \leq 200$.

1.5 Notation, and the names used in the repository [write]

No symbol of the manuscript has been renamed. The table records the letters that carry more than one meaning here, or a different meaning in the Lean developments of PROVEIT, with the tempting false reading beside the true one.

Symbol	Meaning in this article	Remark; tempting false reading
$\theta(n)$	number of 3AP-free permutations of $[n]$ (OEIS A003407)	equals Lean's <code>Sharma2012.theta n</code> and the <code>M n</code> of the Davis–Entringer–Graham–Simmons and LeSaulnier–Vijay developments (<code>Sharma2012.theta_eq_davis_count</code>)
$M(x)$	cutoff-dependent limit of a Cesàro mean, $1 \leq x \leq 2$ (Theorem 7.3 only)	not the count $M(n) = \theta(n)$ of the Lean developments and of [1, 3]
$R(x)$	piecewise-affine interpolation of the toll r_n , (3.2)	not the certificate ratios $R_\pm$ of Theorem 6.4
$C, C_f, C(S)$	$\int_1^2 \Phi$; $\int_1^2 f \circ \Phi$; the subset-state continuation count (10.2)	three different objects
Q	bounded correction in (3.4), values in $[\log 2, \log 21]$	not the toll r_n (whose interpolation is R)
L, U, D	the constants $\log 2, \log 21, \log(21/2)$	not $\liminf b_n$ and $\limsup b_n$ (those are $p_\pm$)
γ	$\log_2(21/2)$ in (9.3)	an exponent, not a growth rate

Unqualified logarithms are natural, and $\log_2$ is the binary logarithm.

2 Combinatorial input and the bounded toll

Definition 2.1. A permutation $\pi = (\pi_1, \dots, \pi_n)$ of $[n] = \{1, \dots, n\}$ is *3AP-free* if there are no indices $i < j < k$ such that $\pi_i + \pi_k = 2\pi_j$. Let $\theta(n)$ be the number of these permutations, with $\theta(0) = 1$. Both increasing and decreasing arithmetic progressions are forbidden; a progression need not occupy consecutive positions.

For example, $\theta(1) = 1$, $\theta(2) = 2$, $\theta(3) = 4$, and $\theta(4) = 10$. The distinction between consecutive positions and arbitrary subsequences is essential to the count.

Lemma 2.2 (Parity splitting). *For every integer $n \geq 2$,*

$$\theta(n) \geq 2\theta(\lfloor n/2 \rfloor)\theta(\lceil n/2 \rceil). \quad (2.1)$$

Proof. Choose a 3AP-free ordering of the even numbers and one of the odd numbers in $[n]$, obtained by affine rescaling of smaller 3AP-free permutations. Concatenate the two blocks, in either order. The two endpoints of any integer three-term progression have the same parity. If its endpoints were in one block and its midpoint in the other, the midpoint could not lie between the endpoints in the concatenated order. If all three entries were in one block, that block's avoidance property would be violated. Thus both concatenations avoid three-term progressions. Their orders and their restrictions to the two nonempty parity blocks determine the input pair, so the construction counts exactly twice the number of input pairs. $\square$

Theorem 2.3 (Sharma's upper splitting bound). *For $n \geq 3$,*

$$\theta(n) \leq 21\theta(\lfloor n/2 \rfloor)\theta(\lceil n/2 \rceil). \quad (2.2)$$

The same inequality is valid at $n = 2$ by direct evaluation.

This is [2, Theorem 2.8]. The proof uses structural restrictions on how fixed odd and even traces can interleave. We do not replace that combinatorial theorem by a numerical fit or an unverified recurrence.

Remark 2.4 (Formal status of the two inputs [write]). Inequality (2.2) for $n \geq 3$ is the Lean theorem `LeanProofs.Sharma2012.theorem_2_8_holds`; the case $n = 2$ is the direct evaluation $\theta(2) = 2 \leq 21$, which the Lean statement does not include. Inequality (2.1) for all $n \geq 2$ is the pair of Lean theorems `even_count_recurrence_holds` ($n = 2k$) and `odd_count_recurrence_holds` ($n = 2k + 1$) in the namespace `LeanProofs.DavisEntringerGrahamSimmons1977`, for $k \geq 1$. The proof of the parity-splitting lemma above is the manuscript's own, and is a second route to a result already machine-checked. See Section 11.

For the remainder of the paper set

$$a_n = \log \theta(n), \quad b_n = \frac{a_n}{n}, \quad L = \log 2, \quad U = \log 21, \quad D = U - L = \log(21/2), \quad (2.3)$$

where unqualified logarithms are natural. Define the *toll*

$$r_n = a_n - a_{\lfloor n/2 \rfloor} - a_{\lceil n/2 \rceil}, \quad n \geq 2. \quad (2.4)$$

Equations (2.1)–(2.2) say precisely that

$$L \leq r_n \leq U. \quad (2.5)$$

This is an exact recurrence with a bounded, generally unknown toll; it is not an assertion that the toll converges or has any simple closed form.

Lemma 2.5 (Elementary global bounds). *For every $n \geq 1$,*

$$(n - 1)L \leq a_n \leq (n - 1)U. \quad (2.6)$$

Proof. The assertion holds at $n = 1$. If it holds at the two smaller arguments in (2.4), their sizes sum to n , and their lower bounds sum to $(n - 2)L$. Adding $r_n \geq L$ gives the lower bound at n ; the upper bound is identical with U in place of L . Strong induction completes the proof. $\square$

3 A continuous growth profile with a bounded correction

3.1 The interpolation identity

Let $A : [1, \infty) \rightarrow \mathbb{R}$ be the piecewise-affine interpolation of a_n :

$$A(n + u) = (1 - u)a_n + ua_{n+1}, \quad 0 \leq u \leq 1. \quad (3.1)$$

For $x \geq 2$, define

$$R(x) = A(x) - 2A(x/2). \quad (3.2)$$

Lemma 3.1 (No interpolation error). *R is the piecewise-affine interpolation of r_n . In particular,*

$$L \leq R(x) \leq U, \quad |R(x) - R(y)| \leq \min\{D|x - y|, D\} \quad (x, y \geq 2). \quad (3.3)$$

Proof. At even integers $n = 2m$ we have $2A(n/2) = 2a_m$; at odd integers $n = 2m + 1$ we have $2A(n/2) = a_m + a_{m+1}$. Thus $R(n) = r_n$ in both cases. On each interval $[n, n + 1]$, both $A(x)$ and $A(x/2)$ are affine: a breakpoint of the latter can only occur at an even integer. Consequently R is affine there, with endpoint values r_n, r_{n+1} . These values lie in $[L, U]$, so its slope has absolute value at most D , and its entire range has diameter at most D . $\square$

3.2 Construction and uniqueness

Theorem 3.2 (Bounded-error log-periodic law). *There is a unique continuous 1-periodic function $P : \mathbb{R} \rightarrow \mathbb{R}$ such that $a_n = nP(\log_2 n) + O(1)$. More precisely, there is a continuous function $Q : [1, \infty) \rightarrow [L, U]$ for which*

$$A(x) = xP(\log_2 x) - Q(x), \quad x \geq 1. \quad (3.4)$$

In particular,

$$\frac{e^{nP(\log_2 n)}}{21} \leq \theta(n) \leq \frac{e^{nP(\log_2 n)}}{2}, \quad n \geq 1. \quad (3.5)$$

Also $L \leq P(t) \leq U$ for every t .

Proof. For $x \geq 1$ and $k \geq 0$ put $F_k(x) = 2^{-k}A(2^k x)$. By (3.2),

$$F_{k+1}(x) - F_k(x) = 2^{-(k+1)}R(2^{k+1}x). \quad (3.6)$$

The uniform bound on R makes this a uniformly convergent telescoping series, indeed uniformly on all $x \geq 1$. Its limit is

$$F(x) = A(x) + \sum_{j=1}^{\infty} 2^{-j}R(2^j x). \quad (3.7)$$

Each summand is continuous, so F is continuous. The tail estimate is

$$L2^{-k} \leq F(x) - F_k(x) \leq U2^{-k}. \quad (3.8)$$

Shifting the defining limit by one index gives $F(2x) = 2F(x)$.

Define $P(t) = 2^{-t}F(2^t)$ initially for $t \geq 0$. The scaling identity gives $P(t+1) = P(t)$ and matching endpoint values, so this defines a continuous 1-periodic function on all of $\mathbb{R}$. Taking

$$Q(x) = F(x) - A(x) = \sum_{j \geq 1} 2^{-j}R(2^j x) \quad (3.9)$$

gives $L \leq Q(x) \leq U$ and proves (3.4). Exponentiation at integer $x = n$ gives (3.5). Interpolating (2.6) and taking the scaled limit gives $Lx \leq F(x) \leq Ux$, which proves the asserted bounds on P .

For uniqueness, suppose a second continuous 1-periodic function P_1 satisfies the same bounded-error assertion. Fix $t \in [0, 1]$ and take $n_k = \lfloor 2^{k+t} \rfloor$. Then $\log_2 n_k - k \rightarrow t$. Division of both asymptotic formulas by n_k and continuity imply $P(t) = \lim b_{n_k} = P_1(t)$. Periodicity extends this equality to every real argument. $\square$

The theorem is the equal-weight bounded-toll case of the framework in [6]; the proof above makes the constants needed here explicit. The bound on Q is a genuine uniform multiplicative control on $\theta(n)$, not just a relative $o(1)$ error in its logarithm. It does *not* assert that $Q(n)$ converges, so it should not be rewritten as an asymptotic equivalence with a fixed prefactor.

3.3 A quantitative modulus of continuity

Write

$$\Phi(s) = \frac{F(s)}{s} = P(\log_2 s), \quad 1 \leq s \leq 2. \quad (3.10)$$

We call s the binary mantissa. The endpoints are identified, since $\Phi(1) = \Phi(2)$.

Proposition 3.3 (Explicit log-Lipschitz control). *For $s, t \in [1, 2]$, $0 < h = |s - t| \leq 1/2$, and $J = \lceil \log_2(1/h) \rceil$,*

$$|\Phi(s) - \Phi(t)| \leq h[L + U + D(J + 1)]. \quad (3.11)$$

Consequently P is Hölder continuous of every exponent less than one, and has a circle modulus $O(h \log(1/h))$ as $h \downarrow 0$.

Proof. On $[1, 2]$, $A(s) = L(s - 1)$. In the series (3.7), each of the first J terms changes by at most Dh when s is replaced by t , by the Lipschitz bound in (3.3). The remaining terms change by at most $D \sum_{j>J} 2^{-j} = D2^{-J} \leq Dh$. Hence $|F(s) - F(t)| \leq Lh + Dh(J + 1)$. Finally,

$$\left| \frac{F(s)}{s} - \frac{F(t)}{t} \right| \leq |F(s) - F(t)| + Uh,$$

because $F(t) \leq Ut$ and $s, t \geq 1$. This proves (3.11). Composition with $s = 2^t$ transfers the estimate to each phase interval. Across the identified endpoints, split the distance at the endpoint and use $\Phi(1) = \Phi(2)$. The Hölder assertions follow from $h \log(1/h) = O_\alpha(h^\alpha)$ for $0 < \alpha < 1$. $\square$

We do not claim that this modulus is optimal for the particular sequence, or that P is differentiable, Lipschitz, or nowhere differentiable.

4 The growth spectrum and finite-data reconstruction

4.1 Every intermediate growth rate occurs

Define

$$p_- = \min_{t \in [0,1]} P(t), \quad p_+ = \max_{t \in [0,1]} P(t), \quad \rho_- = e^{p_-}, \quad \rho_+ = e^{p_+}. \quad (4.1)$$

Theorem 4.1 (Complete subsequential spectrum). *The sets of subsequential limits are*

$$\text{Clust}(b_n) = [p_-, p_+], \quad \text{Clust}(\theta(n)^{1/n}) = [\rho_-, \rho_+]. \quad (4.2)$$

For each $s \in [1, 2]$,

$$\frac{\log \theta(\lfloor 2^k s \rfloor)}{\lfloor 2^k s \rfloor} \longrightarrow \Phi(s), \quad (4.3)$$

with error $O(k2^{-k})$. At an exact integer dyadic ray $n = m2^k$, the error relative to $P(\log_2 m)$ is at most $U/(m2^k)$.

Proof. The identity in Theorem 3.2 gives $|b_n - P(\log_2 n)| \leq U/n$. Every sequence of phases has a convergent subsequence on the circle, so every cluster value belongs to the range of P . Conversely, $\log_2 \lfloor 2^k s \rfloor - k \rightarrow \log_2 s$, proving (4.3). The rounding error in the phase is $O(2^{-k})$; Proposition 3.3 turns it into $O(k2^{-k})$ in the value. A continuous real function on the circle has a compact interval as its range. Exponentiation gives the second statement of (4.2). On an exact dyadic ray there is no phase rounding, giving the stronger error bound directly. $\square$

Thus the failure of a single growth constant is organized by a continuum of phases, rather than by two isolated exceptional subsequences. The exact-period certificate in the next section will in particular show $p_- < p_+$.

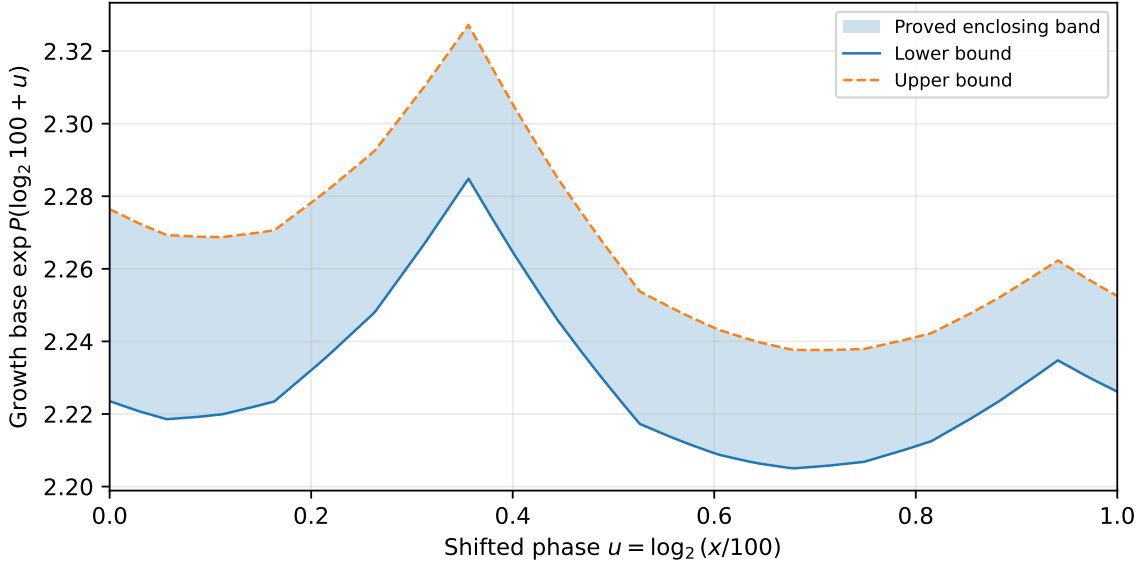


Figure 1: A proved enclosing band from the published counts $100 \leq n \leq 200$. The horizontal coordinate is shifted phase $u = \log_2(x/100)$, not the unshifted phase $\log_2 x$. Both boundaries come from (4.4); no curve is claimed to be the exact limiting profile. Plotting uses floating-point arithmetic, whereas the endpoint and period certificates use exact integers.

4.2 A certified enclosure from one dyadic-width band

Theorem 4.2 (Finite-band reconstruction). *Suppose the counts $\theta(N), \dots, \theta(2N)$ are known, where $N \geq 1$ is an integer. For every real $x \in [N, 2N]$,*

$$\frac{A(x) + L}{x} \leq P(\log_2 x) \leq \frac{A(x) + U}{x}. \quad (4.4)$$

The two envelopes are separated by at most D/N , and the interval of phases covered has length one. Furthermore,

$$\min_{N \leq n \leq 2N} \frac{a_n + L}{n} \leq p_- \leq \min_{N \leq n \leq 2N} \frac{a_n + U}{n}, \quad (4.5)$$

$$\max_{N \leq n \leq 2N} \frac{a_n + L}{n} \leq p_+ \leq \max_{N \leq n \leq 2N} \frac{a_n + U}{n}. \quad (4.6)$$

Each displayed enclosure has width at most D/N .

Proof. Equation (4.4) is (3.4) together with $Q \in [L, U]$. On a unit interval, either envelope has the form $\alpha + \beta/x$, so its maximum and minimum occur at endpoints (or it is constant). Thus extrema over the whole band can be taken at integers. Since $\log_2(2N) - \log_2 N = 1$, the band traverses every phase, which proves (4.5)–(4.6). Pointwise separation is $D/x \leq D/N$; taking maxima or minima cannot increase this bound. $\square$

The theorem gives a uniform approximation algorithm, not an efficient count enumeration algorithm. In principle one can compute all needed finite counts by exhaustive search and hence compute P uniformly, and $p_{\pm}$ as real numbers, to any prescribed accuracy. Achieving log-profile error ε requires counts through $O(\varepsilon^{-1})$ by this bound, but obtaining those counts may be very expensive.

4.3 Numerical bounds with integer verification

Applying Theorem 4.2 to $100 \leq n \leq 200$ gives the following endpoint expressions. The index selection itself can be checked without logarithms: compare $B^{1/m}$ and $C^{1/n}$ by comparing B^n and C^m .

Quantity	Certified bounding expression	Decimal approximation
lower bound for ρ_-	$(2\theta(160))^{1/160}$	2.2049986909
upper bound for ρ_-	$(21\theta(162))^{1/162}$	2.2375996902
lower bound for ρ_+	$(2\theta(128))^{1/128}$	2.2848459606
upper bound for ρ_+	$(21\theta(128))^{1/128}$	2.3272067484

In particular, the independently checked rational endpoint inequalities yield

$$2.20499 < \rho_- < 2.23760 < 2.28484 < \rho_+ < 2.32721. \quad (4.7)$$

These are enclosures, not computed exact values of the extrema. Several of the same endpoint expressions occur in Ho's analysis [5]; numerical priority is not claimed. The point of the finite-band theorem is that it certifies the entire profile and gives a systematic error bound, rather than only selected rays.

5 A finite certificate for the exact fundamental period

The previous sections prove that P is 1-periodic. A function with period one could still have period $1/2$, $1/3$, or another smaller period. Excluding this possibility is essential for the classification of *real* scale factors in (1.1).

Lemma 5.1 (Published-data phase gap). *The following inequalities hold:*

$$e^{P(0)} > \frac{57}{25} = 2.28, \quad \sup_{1/4 \leq t \leq 1/2} e^{P(t)} < \frac{9}{4} = 2.25. \quad (5.1)$$

Proof. All arithmetic needed for the proof is contained in the finite certificate

$$2\theta(128) 25^{128} > 57^{128}, \quad (5.2)$$

$$21\theta(n) 4^n < 9^n \quad (152 \leq n \leq 182). \quad (5.3)$$

These 32 strict integer inequalities are verified by `verify.py` from the published data [7]. First, 128 has phase zero, so Theorem 4.2, or (3.4) directly, gives $e^{P(0)} \geq (2\theta(128))^{1/128} > 57/25$.

Next, (5.3) says $a_n + U < n \log(9/4)$ at every integer in $[152, 182]$. Linear interpolation therefore gives $A(x) + U < x \log(9/4)$ throughout that interval. The real interval

$$[128 \cdot 2^{1/4}, 128\sqrt{2}]$$

is contained in $[152, 182]$: the lower inclusion follows from $19^4 < 2 \cdot 16^4$, and the upper from $2 \cdot 128^2 < 182^2$. For $t \in [1/4, 1/2]$, put $x = 128 \cdot 2^t$ and apply the upper envelope. Periodicity gives $P(\log_2 x) = P(7 + t) = P(t)$, proving the second inequality. The bound is uniform and strict because it holds on a compact interval. $\square$

Theorem 5.2 (Exact period group). *The set of periods of P is exactly $\mathbb{Z}$. In particular, its fundamental positive period is one.*

Proof. The phase gap makes P nonconstant. An irrational period τ , together with period one, would make P constant: the multiples of τ modulo one are dense, and continuity propagates the value at any point along that dense orbit. Thus any additional period must be rational.

If p/q in lowest terms is a period, Bézout's identity and period one show that $1/q$ is a period. Suppose $q \geq 2$. There is an integer j with

$$\frac{1}{4} \leq \frac{j}{q} \leq \frac{1}{2}. \quad (5.4)$$

For $q = 2, 3$ take $j = 1$; for $q \geq 4$ take $j = \lceil q/4 \rceil$. Period $1/q$ would imply $P(j/q) = P(0)$, contradicting (5.1). Therefore $q = 1$. Integer periods already follow from the construction, completing the proof. $\square$

The certificate proves substantially more than nonconstancy, and it does so without locating a maximum or minimum. In particular, it does *not* prove that phase zero is a global maximum. It only proves that a whole specified phase interval lies below its value, which is enough to rule out every smaller period at once.

6 Scaling rigidity and exponential gluing obstructions

6.1 Classification of all positive real scales

For $c > 0$ define, for sufficiently large integers n ,

$$T_c(n) = \log \theta(\lfloor cn \rfloor) - c \log \theta(n), \quad \Delta_c(t) = P(t + \log_2 c) - P(t). \quad (6.1)$$

Lemma 6.1 (Uniform phase formula for a scale change). *For each fixed $c > 0$,*

$$T_c(n) = cn\Delta_c(\log_2 n) + O_c(\log n). \quad (6.2)$$

If c is an integer, the remainder can be replaced by $O_c(1)$.

Proof. Let $m = \lfloor cn \rfloor$. We have $m = cn + O(1)$ and $\log_2 m = \log_2 n + \log_2 c + O_c(1/n)$. By Proposition 3.3, changing the phase by this amount changes P by $O_c(\log n/n)$. Insert this into $a_m = mP(\log_2 m) - Q(m)$, subtract $ca_n = cnP(\log_2 n) - cQ(n)$, and use boundedness of P and Q . If c is integral, there is no rounding and the remainder is exactly $cQ(n) - Q(cn)$, hence bounded. $\square$

Theorem 6.2 (Real-scale rigidity and frequent gain/loss). *For $c > 0$, $T_c(n) = o(n)$ if and only if $c = 2^j$ for an integer j . For every other c there are $\eta_c > 0$ and two sets $E_c^+, E_c^- \subseteq \mathbb{N}$ of positive lower natural density such that*

$$\theta(\lfloor cn \rfloor) \geq e^{\eta_c n} \theta(n)^c \quad (n \in E_c^+), \quad (6.3)$$

$$\theta(\lfloor cn \rfloor) \leq e^{-\eta_c n} \theta(n)^c \quad (n \in E_c^-). \quad (6.4)$$

Finite initial exceptions may be removed from these sets.

Proof. If $c = 2^j$, periodicity makes Δ_c zero, and (6.2) proves $T_c = o(n)$. Conversely, if $T_c = o(n)$, apply (6.2) along $n_k = \lfloor 2^{k+t} \rfloor$ for each fixed t . Continuity yields $\Delta_c(t) = 0$. Thus $\log_2 c$ is a period of P , and Theorem 5.2 gives $\log_2 c \in \mathbb{Z}$.

Now take an off-dyadic scale. The continuous periodic function Δ_c is not identically zero and has integral zero on a period, by translation invariance of the integral. It therefore has both positive and negative values. Continuity supplies phase intervals on which $c\Delta_c$ is bounded

below by $2\eta_c$ or above by $-2\eta_c$, for some $\eta_c > 0$. Equation (6.2) gives (6.3) and (6.4) at all sufficiently large integers with those phases.

For completeness, such phase intervals have positive lower natural density. Choose a closed mantissa interval $[u, v] \subset (1, 2)$ of positive length inside one of them. If $2^k \leq N < 2^{k+1}$, the complete dyadic blocks below 2^k contain at least

$$(v - u)(2^k - 1) - O(k)$$

integers with mantissa in $[u, v]$. Division by $N < 2^{k+1}$ gives lower density at least $(v - u)/2 > 0$. Apply this to both sign intervals. $\square$

Corollary 6.3 (Integer arities). *For an integer $m \geq 2$, a uniform subexponential comparison $\theta(mn) = \theta(n)^m \exp(o(n))$ holds exactly when m is a power of two. For $m = 2^r$ the stronger bounds*

$$2^{m-1}\theta(n)^m \leq \theta(mn) \leq 21^{m-1}\theta(n)^m \quad (6.5)$$

hold for every $n \geq 1$.

Proof. The classification is Theorem 6.2. Iterating the even recurrence r times gives factors with total exponent $1 + 2 + \dots + 2^{r-1} = m - 1$, which proves (6.5). $\square$

There is an important quantifier here: the error $o(n)$ must hold as n runs through *all* sufficiently large integers. Particular off-dyadic phases can still have zero scale difference. The theorem does not exclude such accidental equalities of profile values.

6.2 Explicit exponential failure at the split $2n + n$

Define the unequal-block gluing ratio

$$G(n) = \frac{\theta(3n)}{\theta(2n)\theta(n)}. \quad (6.6)$$

The parity recurrence gives $\log \theta(2n) - 2 \log \theta(n) \in [L, U]$, so

$$\log G(n) = 3n\Delta_3(\log_2 n) + O(1). \quad (6.7)$$

In particular, exponential loss and exponential gain for $G(n)$ each occur on a set of positive lower natural density. The following stronger explicit subsequences need only four count values.

Theorem 6.4 (Certified 2:1 gluing loss and gain). *For every integer $t \geq 0$,*

$$G(128 \cdot 2^t) < \left(\frac{49}{50}\right)^{128 \cdot 2^t}, \quad (6.8)$$

$$G(172 \cdot 2^t) > \left(\frac{21}{20}\right)^{172 \cdot 2^t}. \quad (6.9)$$

Proof. Iteration of the even recurrence gives, for $s \geq 0$,

$$2^{2^s-1}\theta(k)^{2^s} \leq \theta(k2^s) \leq 21^{2^s-1}\theta(k)^{2^s}. \quad (6.10)$$

Set

$$R_- = \frac{(21\theta(192))^2}{(2\theta(128))^3}, \quad R_+ = \frac{(2\theta(129))^4}{(21\theta(172))^3}. \quad (6.11)$$

The integer certificates checked by the accompanying program are

$$(21\theta(192))^2 50^{128} < (2\theta(128))^3 49^{128}, \quad (6.12)$$

$$(2\theta(129))^4 20^{172} > (21\theta(172))^3 21^{172}. \quad (6.13)$$

Thus $R_- < (49/50)^{128}$ and $R_+ > (21/20)^{172}$.

For $n = 128 \cdot 2^t$, write $3n = 192 \cdot 2^{t+1}$. Using the upper bound in (6.10) in the numerator and the lower bounds in the denominator of $G(n)$ gives

$$G(n) \leq \frac{4}{21} R_-^{2^t} < (49/50)^n. \quad (6.14)$$

For $n = 172 \cdot 2^t$, write $3n = 129 \cdot 2^{t+2}$. Use the lower bound for the numerator and upper bounds for both denominator factors instead. This gives

$$G(n) \geq \frac{21^2}{2} R_+^{2^t} > (21/20)^n. \quad (6.15)$$

The prefactors are respectively less than and greater than one, so no asymptotic threshold is needed: both statements hold for every $t \geq 0$. $\square$

Corollary 6.5 (No subexponential-loss universal gluing). *There is no function $\varepsilon(N) \rightarrow 0$ such that, for all sufficiently large u, v ,*

$$\theta(u+v) \geq e^{-\varepsilon(u+v)(u+v)} \theta(u) \theta(v). \quad (6.16)$$

Nor does the analogous upper bound with $e^{\varepsilon(u+v)(u+v)}$ hold uniformly for all large u, v .

Proof. Take $u = 2n, v = n$ along the loss or gain subsequence in Theorem 6.4. The proposed error exponent is $o(n)$, whereas the exhibited logarithmic discrepancy has magnitude bounded below by a positive constant times n . $\square$

The loss has a direct counting interpretation. Along the first subsequence, any map from all pairs of 3AP-free permutations of sizes $2n$ and n into 3AP-free permutations of size $3n$ must identify exponentially many pairs somewhere, if it is defined on the entire domain. In particular, a universal injective concatenation-type construction at this size ratio is impossible. This does not rule out successful constructions on a restricted family or methods that record additional output information.

7 Averaging and complete empirical limit laws

The interval of cluster values describes which limits are possible. It does not describe how often the values occur. Because the phase is logarithmic, logarithmic and ordinary sampling answer this question differently.

7.1 Logarithmic averaging has a unique limiting law

Let $H_N = \sum_{n=1}^N 1/n$ and define probability measures

$$\lambda_N = \frac{1}{H_N} \sum_{n=1}^N \frac{1}{n} \delta_{b_n}. \quad (7.1)$$

Here δ_x is the point mass at x , and convergence of probability measures means convergence against every continuous test function on a compact interval containing their supports.

Theorem 7.1 (Logarithmic empirical law). *For every continuous test function f ,*

$$\lim_{N \rightarrow \infty} \frac{1}{H_N} \sum_{n \leq N} \frac{f(b_n)}{n} = \int_0^1 f(P(t)) dt = \frac{1}{\log 2} \int_1^2 f(\Phi(s)) \frac{ds}{s}. \quad (7.2)$$

For Lipschitz f the error is $O_f(1/\log N)$. The limiting measure has support $[p_-, p_+]$. Replacing b_n by $\theta(n)^{1/n}$ pushes this law forward under the exponential map.

Proof. First take f Lipschitz. Replacing b_n by $P(\log_2 n)$ changes the unnormalized weighted sum by at most $\text{Lip}(f)U \sum n^{-2} = O_f(1)$. On $[n, n+1]$, the phase changes by $O(1/n)$, so the modulus of P gives a change $O(\log n/n)$ in its value. The difference between the resulting sum and the integral is therefore bounded by a constant times $\sum_{n \geq 2} \log n/n^2$, together with a bounded initial contribution. Consequently,

$$\sum_{n \leq N} \frac{f(b_n)}{n} = \int_1^N f(P(\log_2 x)) \frac{dx}{x} + O_f(1) \quad (7.3)$$

$$= (\log N) \int_0^1 f(P(t)) dt + O_f(1), \quad (7.4)$$

where the last step splits the logarithmic integral into complete periods and one bounded remainder. Since $H_N = \log N + O(1)$, this proves the asserted rate. Continuous test functions follow by uniform approximation by piecewise-affine, hence Lipschitz, functions on the compact range.

Every neighborhood of a value in the image of a continuous P has a preimage containing a nonempty phase interval, of positive Lebesgue measure. Thus the support is the entire image $[p_-, p_+]$. The substitution $s = 2^t$ proves the mantissa formula, and continuity of the exponential gives the final assertion. $\square$

7.2 Ordinary cutoffs retain a phase

Define the ordinary empirical measure

$$\mu_N = \frac{1}{N} \sum_{n=1}^N \delta_{b_n}. \quad (7.5)$$

For $x \in [1, 2]$ define a probability measure ν_x by its test-function integrals:

$$\begin{aligned} \nu_x(f) &= \frac{1}{x} \left(\int_1^2 f(\Phi(s)) ds + \int_1^x f(\Phi(s)) ds \right) \\ &= \frac{2}{x} \int_1^x f(\Phi(s)) ds + \frac{1}{x} \int_x^2 f(\Phi(s)) ds. \end{aligned} \quad (7.6)$$

The weights integrate to one. At the endpoints $\nu_1 = \nu_2$, as required when the cutoff phase wraps around.

Theorem 7.2 (All ordinary empirical limit laws). *For $N_k = \lfloor 2^k x \rfloor$ with $x \in [1, 2]$,*

$$\mu_{N_k} \implies \nu_x. \quad (7.7)$$

More generally, the complete set of weak subsequential limits of μ_N is $\{\nu_x : 1 \leq x \leq 2\}$. The map $x \mapsto \nu_x$ is weakly continuous and all its measures have support $[p_-, p_+]$.

Proof. For Lipschitz f , replacing b_n by $P(\log_2 n)$ changes its ordinary average by $O_f(H_N/N)$, which tends to zero. Continuous f again follow by uniform approximation. Put $u_n = n/2^k$. Since P is 1-periodic, $P(\log_2 n) = P(\log_2 u_n)$. Riemann sums on $[\varepsilon, x]$, followed by $\varepsilon \downarrow 0$, give

$$\lim_{k \rightarrow \infty} \mu_{\lfloor 2^k x \rfloor}(f) = \frac{1}{x} \int_0^x f(P(\log_2 u)) du. \quad (7.8)$$

The integrand is bounded and continuous away from zero, so the omitted interval has arbitrarily small contribution.

Let $C_f = \int_1^2 f(\Phi(s)) ds$. Splitting $(0, 1]$ into dyadic intervals and substituting $u = 2^{-j}s$ shows

$$\int_0^1 f(P(\log_2 u)) du = \sum_{j \geq 1} 2^{-j} C_f = C_f. \quad (7.9)$$

Thus (7.8) equals (7.6).

The same Riemann-sum argument holds for a sequence of upper endpoints $x_k \rightarrow x$, by boundedness of the integrand and the small length of the interval between x_k and x . Every sequence of cutoffs has a subsequence whose mantissas $N/2^{\lfloor \log_2 N \rfloor}$ converge in $[1, 2]$. This proves that no other weak limit is possible, while the fixed-mantissa cutoffs realize every ν_x . Formula (7.6) proves weak continuity. Its mantissa density is at least $1/2$ throughout $[1, 2]$, so the support is the full image of Φ , by the same neighborhood argument used for the logarithmic law. $\square$

Theorem 7.3 (Ordinary averages do not wash out the oscillation). *Neither of the ordinary Cesàro averages*

$$\frac{1}{N} \sum_{n \leq N} \frac{\log \theta(n)}{n}, \quad \frac{1}{N} \sum_{n \leq N} \theta(n)^{1/n} \quad (7.10)$$

converges. In particular, neither corresponding ordinary empirical measure has a single weak limit.

Proof. For the first average its cutoff-dependent limit is

$$M(x) = \frac{C + \int_1^x \Phi(s) ds}{x}, \quad C = \int_1^2 \Phi(s) ds. \quad (7.11)$$

Continuity of Φ makes M continuously differentiable, with

$$M'(x) = \frac{\Phi(x) - M(x)}{x}. \quad (7.12)$$

If M were constant, (7.12) would force Φ to be constant, contradicting the phase gap. Thus two cutoff phases give distinct limiting means. For the second average replace Φ throughout by e^Φ , which is likewise continuous and nonconstant. Nonconvergence of these continuous test-function integrals precludes weak convergence of the empirical measures. $\square$

Remark 7.4. Full support is not the same as existence of a probability density. The pushforward laws could in principle have atoms at values corresponding to positive-measure level sets of P . Nothing in the arguments above proves absolute continuity, absence of atoms, or a particular shape of a histogram.

8 Polynomial and geometric sampling

8.1 Multiplicatively fine sampling preserves the spectrum

Proposition 8.1. *Let n_k be a strictly increasing unbounded sequence of positive integers with $n_{k+1}/n_k \rightarrow 1$. Then*

$$\text{Clust}(b_{n_k}) = [p_-, p_+], \quad \text{Clust}(\theta(n_k)^{1/n_k}) = [\rho_-, \rho_+]. \quad (8.1)$$

In particular this holds along every eventually positive arithmetic progression with positive common difference and every eventually positive, nonconstant integer-valued polynomial sampled at positive integers.

Proof. Fix $s \in [1, 2]$ and let $n_{k(j)}$ be the first sample not below $2^j s$. Its predecessor is below $2^j s$, so

$$1 \leq \frac{n_{k(j)}}{2^j s} \leq \frac{n_{k(j)}}{n_{k(j)-1}} \rightarrow 1.$$

Consequently its phases converge to $\log_2 s$ modulo one, and the profile law gives limit $\Phi(s)$. These values fill the interval; no new values are possible because the sampled sequence is a subsequence of the original. A polynomial of positive degree and positive leading coefficient is strictly increasing eventually and has successive ratios tending to one. Arithmetic progressions are the degree-one case. Removing a finite initial segment changes none of the assertions. $\square$

This assertion concerns the cluster set, not the frequency distribution under polynomial sampling. It already shows that the continuum of growth rates is not confined to specially chosen powers of two.

8.2 Geometric sampling is governed by rotation

Theorem 8.2 (Geometric sampling classification). *Let $c > 0$, $b > 1$, $n_k = \lfloor cb^k \rfloor$, and put $\alpha = \log_2 b$, $\beta = \log_2 c$. Ignore any initial zero terms.*

(i) *If $\alpha = p/q$ is rational in lowest terms, the cluster values of b_{n_k} are*

$$\{P(\beta + jp/q) : 0 \leq j < q\}, \quad (8.2)$$

and the empirical law is the average of the q corresponding point masses. Some of the displayed values may coincide.

(ii) *If α is irrational, the cluster set is $[p_-, p_+]$, and*

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K f(b_{n_k}) = \int_0^1 f(P(t)) dt \quad (8.3)$$

for every continuous f .

Both statements transfer to $\theta(n_k)^{1/n_k}$ by exponentiation.

Proof. Rounding gives $\log_2 n_k = \beta + k\alpha + O(b^{-k})$. The modulus of continuity and bounded correction imply

$$b_{n_k} = P(\beta + k\alpha) + O(kb^{-k}). \quad (8.4)$$

In the rational case the phase sequence is periodic with period q , and each residue class of k has frequency $1/q$.

For irrational α , each nonzero integer h satisfies

$$\frac{1}{K} \sum_{k=1}^K e^{2\pi i h(\beta + k\alpha)} \longrightarrow 0,$$

by the finite geometric-sum formula, because $e^{2\pi i h\alpha} \neq 1$. Approximation of continuous periodic functions by trigonometric polynomials gives uniform distribution of these phases. Apply it to $f \circ P$. The orbit is dense, so continuity also gives the full cluster interval. The vanishing rounding error changes neither the cluster sets nor the limiting averages. $\square$

For example, sampling $n = 3^k$ produces the full growth interval and the logarithmic limiting law, because $\log_2 3$ is irrational. In contrast, sampling $n = 2^k$ converges to the single phase-zero growth value. Exact fundamental period one does not mean that different phases always have different values: a continuous real periodic function cannot be injective on its circle.

9 Local ratios and the remaining monotonicity problem

The profile theorem should not be confused with a proof that $\theta(n)$ increases at every step. Ho explicitly distinguishes that open question from the root-growth question [5]. The following exact identity shows where the difficulty resides.

Proposition 9.1 (Binary descent for adjacent logarithmic ratios). *Let $d_n = a_{n+1} - a_n$. Then $d_1 = L$, and for $n \geq 2$,*

$$d_n = d_{\lfloor n/2 \rfloor} + r_{n+1} - r_n. \quad (9.1)$$

Consequently,

$$L - D \lfloor \log_2 n \rfloor \leq d_n \leq L + D \lfloor \log_2 n \rfloor, \quad (9.2)$$

and, with $\gamma = \log_2(21/2)$,

$$2n^{-\gamma} \leq \frac{\theta(n+1)}{\theta(n)} \leq 2n^\gamma. \quad (9.3)$$

The elementary deletion bound improves the upper side to $n+1$.

Proof. For $n = 2m$, subtract the recurrences for a_{2m+1} and a_{2m} ; for $n = 2m+1$, subtract those for a_{2m+2} and a_{2m+1} . Both subtractions produce (9.1). Iteration reaches index one after exactly $\lfloor \log_2 n \rfloor$ steps. Each toll difference is in $[-D, D]$, proving (9.2); exponentiation gives (9.3). For the deletion bound, remove the entry $n+1$ from a 3AP-free permutation of $[n+1]$. The remaining ordering avoids all progressions in $[n]$. Together with its original insertion position, this is an injective encoding into at most $(n+1)\theta(n)$ possibilities. $\square$

The lower bound in (9.3) is positive but falls below one; it therefore does not solve monotonicity. Proving $d_n \geq 0$ would require more information about the correlated toll differences along binary descent than the independent bounds $r_n \in [L, U]$ supply.

The nondegenerate growth interval also prevents eventual log-convexity or eventual log-concavity. Either property would make adjacent logarithmic ratios eventually monotone. The exponential bounds on a_n would then force their Cesàro means a_n/n to converge, contradicting Theorem 4.1. This is a limitation of possible regularity arguments, not a substitute for the unresolved monotonicity question.

Remark 9.2 (Adjacent ratios have no limit [write]). The same Cesàro argument shows that $\theta(n+1)/\theta(n)$ has no limit in $[0, \infty]$. A finite positive limit λ would give $d_n \rightarrow \log \lambda$ and hence $b_n \rightarrow \log \lambda$, contradicting Theorem 4.1 with $p_- < p_+$ (or Ho's theorem [5]); the limits 0 and

∞ would force $b_n \rightarrow -\infty$ and $b_n \rightarrow +\infty$ respectively, contradicting (2.6). The special case that the ratio does not tend to 2 is machine-checked in PROVEIT (Section 11); the general statement is not formalized. The upper side of Proposition 9.1 is also improved in Lean, to $\theta(n+1) \leq \lfloor (n+3)/2 \rfloor \theta(n)$.

[Added 29 September 2026, batch 43: the deletion bound at the end of Proposition 9.1 is the one local estimate that Part II needs (Lemma 17.1). Together with Ho’s nonconvergence it shows that $\theta(n)$ is not P-recursive (Theorem 16.2); the binary-descent identity is reused there as Lemma 17.4.]

10 Reproducible verification and a formalization route

10.1 What the package checks

The program `verify.py` uses only the Python standard library. Its executed record is `verification.json`. It performs the following logically distinct tasks.

1. It checks all 199 lower and upper splitting inequalities in the supplied range $2 \leq n \leq 200$. These are consistency checks on a finite table, not a proof of the infinite recurrence.
2. It checks the 32 phase-gap inequalities, the two interval-containment inequalities, the two explicit gluing certificates, and two previously used Ho separation certificates, with exact integer arithmetic.
3. It chooses the extremizing indices in each band $[32, 64]$, $[64, 128]$, and $[100, 200]$ by exact power comparisons. It encloses each selected root between decimal rationals by integer bisection. Decimal logarithms are used only to print additional approximations.
4. It independently recomputes $\theta(n)$ for every $0 \leq n \leq 32$ using a subset-state recurrence, and cross-checks $0 \leq n \leq 8$ by direct permutation enumeration.

All these executed checks passed. No claim is made that the independent Python enumeration was run for $33 \leq n \leq 200$, or that a Python assertion is a proof checked by a theorem prover.

[write] In the collection the program is shipped as `code/verify.py` and its executed record as `data/verification.json`; the counts are `data/counts.txt`. The program locates its data relative to its own directory, so it must be run from a copy that restores the delivered flat layout (program beside `data/`); the report’s README gives the commands. A rerun of `verify.py -max-n 32` on such a copy, made when the report was written, passed and reproduced the shipped record except for its timing fields.

10.2 Why the independent enumeration is correct

For a fixed n , represent the set of values already placed in a prefix by $S \subseteq [n]$. A candidate next value $y \notin S$ is allowed if, for every $d \geq 1$ with $y-d, y+d \in [n]$, either both endpoints belong to S or neither does. Equivalently,

$$\mathbf{1}_S(y-d) = \mathbf{1}_S(y+d). \quad (10.1)$$

Let $C(S)$ be the number of continuations generated by this rule, with $C([n]) = 1$ and

$$C(S) = \sum_{\substack{y \notin S \\ y \text{ allowed}}} C(S \cup \{y\}). \quad (10.2)$$

Then $C(\emptyset) = \theta(n)$.

To prove this, consider a complete permutation and a prospective middle value y of an arithmetic progression. At the moment y is placed, that progression appears as a subsequence if and only if exactly one of its two endpoints has already appeared. Thus (10.1) excludes exactly the forbidden full permutations. Every full permutation corresponds to exactly one chain of subsets and choices, so (10.2) counts each permitted permutation once. The condition depends only on the used set, which justifies memoization; no unrecorded ordering information is needed for this rule. This is an independently written implementation of the standard subset-state idea, not a claim of a new enumeration algorithm [4].

10.3 Certificate design for a proof assistant

A practical formalization can be layered rather than beginning with the analytic limit.

First prove the finite permutation definition, the parity injection, and soundness of (10.2). Next expose Sharma’s estimate as an explicit imported theorem or an explicitly named hypothesis until its full combinatorial proof has been formalized. Define a_n and r_n , prove the piecewise-affine interpolation lemma, and construct F by uniform convergence of the geometric tail. This supplies the periodic profile without any numerical experiments.

The period and gluing proofs then reduce to a small finite certificate over natural-number powers, plus the already proved interpolation envelope. A verified certificate evaluator should distinguish “the table entry is the count” from “the inequality involving that entry is true.” Finally, formalize the density and integral arguments on compact intervals. This separation prevents a conditional analytic theorem from being reported as an unconditional theorem about the combinatorial sequence.

The present package contains no Lean file asserting these results. The route just described is an implementation plan, not an executed formalization.

[write] Part of the first layer already exists in PROVEIt, and the second layer’s “imported theorem” is not needed there: the permutation definition, the parity construction and both parity recurrences, the count identification between the three papers’ definitions, and Sharma’s Theorem 2.8 itself are proved in Lean (Section 11). The soundness of the subset-state recurrence (10.2), the analytic profile, and every later layer are not formalized. The existing finite-count evaluator stops at $n = 20$ and runs through `native_decide` on a native backend, which is itself a trust boundary; the certificates above need n between 100 and 200.

11 Formal status in ProveIt [write]

This section was added when the report entered the collection. It records what PROVEIt’s Lean development of the three source papers [1, 2, 3] already proves about the objects of this article, and what it does not. (The Lean namespace `Sharma2012` formalizes the paper cited here as [2], 2009.) The Lean files are under `Combinatorics/Ramsey/Lean/`; the paths below are relative to that directory, the namespaces begin with `LeanProofs.`, and line numbers are those of the current tree, which agrees with the manuscript’s pinned revision in this directory. No Lean build and no axiom audit were run for this report.

11.1 Declarations that prove inputs of this article

- `Sharma2012.theorem_2_8_holds` (`Sharma2012/FinalConsequences.lean:23`), proving `Sharma2012.theorem_2_8` (`Sharma2012/Statements.lean:272`):

$\theta(n) \leq 21 \theta(\lfloor (n+1)/2 \rfloor) \theta(\lfloor n/2 \rfloor)$ for $n \geq 3$. This is (2.2), the one theorem the manuscript imports. The case $n = 2$ is not part of the Lean statement.

- `DavisEntringerGrahamSimmons1977.even_count_recurrence_holds` and `odd_count_recurrence_holds`
(`DavisEntringerGrahamSimmons1977/CountingRecurrences.lean`, lines 252 and 485): $2M(k)^2 \leq M(2k)$ and $2M(k+1)M(k) \leq M(2k+1)$ for $k \geq 1$. Together they are (2.1) for every $n \geq 2$. The same inequalities are restated for Sharma’s θ as `Sharma2012.inequality_1_holds` and `inequality_2_holds`
(`Sharma2012/CountingConsequences.lean`, lines 19 and 24) and for LeSaulnier–Vijay’s M as `LeSaulnierVijay2011.M_even_recurrence_holds` and `M_odd_recurrence_holds`
(`LeSaulnierVijay2011/Proofs.lean`, lines 94 and 100).
- `Sharma2012.theta_eq_davis_count` (`Sharma2012/AsymptoticConsequences.lean:22`): Sharma’s θ and the Davis–Entringer–Graham–Simmons M are the same count.
- `Sharma2012.theorem_1_1_holds` and `DavisEntringerGrahamSimmons1977.fact_1_holds`: $2^{n-1} \leq \theta(n)$ for $n \geq 1$, the lower half of (2.6). The upper half, $\theta(n) \leq 21^{n-1}$, is not stated in Lean; Sharma’s `theorem_2_9_holds` proves the different bound $\theta(n) \leq (27/10)^n/21$ for $n \geq 11$.
- `DavisEntringerGrahamSimmons1977.table_1_holds`: the values $\theta(1), \dots, \theta(20)$, which agree with the first twenty entries of `data/counts.txt`. They rest on `RamseyPaperCommon.reflectedM_values_through_twenty`
(`RamseyPaperCommon/CountCertificates.lean:17`), proved by `native_decide` through an `implemented_by` native backend (`RamseyPaperCommon/ThreeFreeCounting.lean`, lines 359–374). These values therefore carry that explicit trust boundary. Whether the declarations above depend on it was not audited here; the import closure of `Sharma2012/FinalConsequences.lean` contains it.

11.2 Declarations bearing on the article’s discussion

- `DavisEntringerGrahamSimmons1977.insertion_count_upper_bound_holds`
(`DavisEntringerGrahamSimmons1977/Proofs.lean:4824`): $\theta(n+1) \leq \lfloor (n+3)/2 \rfloor \theta(n)$ for $n \geq 1$. This is stronger than the deletion bound $(n+1)\theta(n)$ at the end of Proposition 9.1.
- `LeSaulnierVijay2011.M_ratio_does_not_tend_to_two_holds`
(`LeSaulnierVijay2011/Proofs.lean:231`): $\theta(n+1)/\theta(n)$ does not tend to 2. This is the negative answer to the question Sharma records as `Sharma2012.introduction_ratio_question`. Remark 9.2 below derives the stronger unformalized statement that the ratio has no limit at all.
- `LeSaulnierVijay2011.theorem_1_holds`: $\theta(n) \geq \frac{1}{2} 2132^{n/10}$ for $n \geq 8$, and Sharma’s `theorem_2_10_holds`: $\limsup \theta(n)^{1/n} \leq 27/10$. These are the formal exponential bounds. The enclosure (4.7) is narrower but rests on published counts that PROVEIt does not certify.
- `Sharma2012.open_problem_1` (monotonicity of θ) and `open_problem_2` (convergence of $\theta(n)^{1/n}$) (`Sharma2012/Statements.lean`, lines 312 and 316) have no proof declarations, by design (`Sharma2012/Proofs.lean`, lines 6–9). The first is still open, as Section 9 says. The second is answered negatively by Ho’s theorem [5], and again by Theorems 4.1 and 5.2; neither answer is formalized, and the Lean docstring correctly describes the question as

open in Sharma’s paper. `open_problem_3`, whether $\theta(n+1) < 3\theta(n)$ always, is untouched by this article.

11.3 What is not formalized

None of the article’s own results is formalized: the bounded-toll profile (Theorem 3.2), its modulus, the spectrum and band theorems, the exact period (Theorem 5.2), the real-scale classification (Theorem 6.2), the explicit gluing certificates, the empirical and sampling laws, the binary-descent identity, and the soundness of the subset-state enumeration. The integer certificates are checked by a Python program only, from published counts that PROVEIT does not certify. Placement of this report beside the Lean development gives none of these statements formal status.

11.4 Part II, added in batch 43

[Added 29 September 2026.] Part II was checked against the same Lean files. They are unchanged under `Combinatorics/Ramsey/Lean/` between Part I’s pin 74f7f5bdb, Part II’s pin 8d936ee23 and the current tree, so the names, paths and line numbers above apply. Again no Lean build and no axiom audit were run.

Lean theorems whose statements Part II uses.

- `Sharma2012.theorem_2_8_holds`, the upper splitting bound for $n \geq 3$ (Proposition 17.2; its case $n = 2$ is the evaluation $\theta(2) = 2$). Part II relies on this statement twice: through $\theta(n) \leq 21^{n-1}$ (Lemma 17.3), which is the exponential bound that the G-function input needs, and through the factor 21 in Ho’s separation $(2\theta(64))^{75} > (21\theta(75))^{64}$, which enters as the upper enclosure of ℓ_{75} in Lemma 18.1. The manuscript imports the published theorem and does not cite the Lean proof as evidence; the Lean proof is a formal proof of that statement, not of anything Part II derives from it.
- `DavisEntringerGrahamSimmons1977.even_count_recurrence_holds` and `odd_count_recurrence_holds`, the lower splitting bound (the lower halves of Proposition 17.2 and Lemma 17.3, and the lower enclosure of ℓ_{64}); `Sharma2012.theorem_1_1_holds`, $2^{n-1} \leq \theta(n)$; and `Sharma2012.theta_eq_davis_count`, which identifies the counts.

A stronger Lean form of Part II’s local estimate. `DavisEntringerGrahamSimmons1977.insertion_count_upper_bound_holds` proves $\theta(n+1) \leq \lfloor (n+3)/2 \rfloor \theta(n)$ for $n \geq 1$. This is stronger than the deletion bound $\theta(n+1) \leq (n+1)\theta(n)$ of Lemma 17.1: the two agree at $n = 1$, and the Lean factor is strictly smaller for every $n \geq 2$. Iterating it gives the window bound (17.2) as well. Part II needs only some polynomial bound on $\theta(n+1)/\theta(n)$, so nothing in it changes; the point is that the one local estimate its main theorem uses has a machine-checked, sharper counterpart in PROVEIT.

Counts that nothing in ProveIt certifies. The two counts $\theta(64)$ and $\theta(75)$ of (18.3), from which Part II reproduces Ho’s nonconvergence (Theorem 18.2) and on which its theorems about θ therefore rest, are published counts (Ho; Correll–Ho; OEIS). The general lemmas of Section 19 do not use them. No Lean or Rocq development in PROVEIT certifies them, and no Lean declaration mentions them: the Lean count table `table_1_holds` stops at $\theta(1), \dots, \theta(20)$, behind the `native_decide/implemented_by` trust boundary described above. Part II’s Python verifier recomputes $\theta(n)$ only for $n \leq 32$ and checks the integer comparison, not the counts. The two values agree with rows 64 and 75 of Part I’s `data/counts.txt`

(checked when Part II was written), but that file holds the same published OEIS data, so the agreement is not an independent confirmation. Part I's own route to $p_- < p_+$ uses other published counts, equally uncertified.

Open problems. Ho's theorem, reproduced as Theorem 18.2, is again the negative answer to Sharma2012.open_problem_2; it is not formalized. Part II does not decide open_problem_1 (monotonicity; Question 26.4). The Ramsey development states nothing about P-recursiveness or D-finiteness.

What is not formalized. None of Part II's results is formalized: rational descent (Lemma 19.1), the imported G-function input (Theorem 19.2), finite-window noncancellation, local-growth rigidity and the jump corollary, the section-limit lemma, the main theorem, robustness, rounded powers, positive polynomial observables, the Nilsson-layer exclusion, the generating-function and transcendence corollaries, the profile obstruction, perturbed sampling, the soundness of the subset-state enumeration, and the two determinant certificates, which a Python program checks. The deep imported input, the arithmetic theory of G-functions, has no counterpart in PROVEIT. The manuscript's formalization plan (Section 25) is a plan, not an executed formalization. Placement of Part II beside the Lean development gives it no formal status.

12 Further research questions and topics

The results narrow several questions without resolving all of the fine structure. The following proposed program separates concrete next steps from more speculative extensions.

1. **Extremizing phases and sharp constants.** Determine the exact values or substantially narrower enclosures for $p_{\pm}$, and characterize their maximizing and minimizing phase sets. The certificate used here does not show that $P(0)$ is a global maximum, nor that either extremum has a unique phase. Larger count bands immediately produce better certified enclosures, while a structural argument could be far more informative than additional enumeration.
2. **Regularity beyond the general bound.** Is P Lipschitz? Is it differentiable at typical phases, at any dyadic phase, or on a set of full measure? Is the logarithmic loss in (3.11) actually necessary? These questions require information about cancellations between toll increments; boundedness alone does not settle them.
3. **The distributions of growth values.** Determine whether the logarithmic pushforward law has atoms, a density, or a singular component. Study the dependence of the ordinary laws ν_x on x , including when two distinct cutoff phases produce the same law. The formula (7.6) is exact, but it does not determine the regularity of the pushforward without further knowledge of level sets of P .
4. **The bounded multiplicative correction.** Understand $Q(n)$ and the tolls r_n along dyadic and other arithmetic subsequences. A limiting correction would strengthen a phase-dependent exponential envelope to an asymptotic equivalence on that subsequence. Nothing proved here implies such convergence, and a nontrivial limiting law for the corrections may be more realistic than a constant prefactor.
5. **Monotonicity and adjacent ratios.** Decide whether $\theta(n+1) \geq \theta(n)$ always holds. The binary-descent identity (9.1) offers a precise target: control a sum of correlated toll differences, not merely each difference separately. A combinatorial injection would resolve

monotonicity but must accommodate the unequal-size exponential obstructions proved in this paper. [write] This is Sharma’s open problem (1), recorded without proof in PROVEIT as `Sharma2012.open_problem_1`.

- 6. Sharper interleaving bounds and effective computation.** Can the uniform factor 21 be reduced, perhaps beyond a finite threshold, or replaced by a phase-sensitive bound? Either improvement would sharpen all the enclosures automatically. Separately, develop an independently checkable enumeration pipeline for larger count bands with explicit memory, time, and certificate-size accounting.
- 7. Machine-checked end-to-end certificates.** Formalize the complete implication from a definition of θ and a verified finite-count evaluator to the phase gap, exact period, and gluing inequalities. The analytic construction is reusable for other balanced recurrences. The main proof-engineering challenge is avoiding hidden large count assumptions while keeping the certificate small. [write] In PROVEIT the definition of θ , both splitting recurrences (including Sharma’s upper one) and the values through $n = 20$ are already formalized (Section 11). What remains is the verified evaluator for the counts the certificates use ($100 \leq n \leq 200$) without a native trust boundary, the analytic profile, and the certificate implications themselves.
- 8. Other avoidance classes and generating functions.** Search for analogous bounded-toll recurrences in longer-progression avoidance or other permutation classes. Ask whether the resulting profile can have a smaller fundamental period or whether an exact scale group can again be read from finite data. For $\sum \theta(n)z^n$, investigate what the nonconstant exponential profile implies about analytic continuation or algebraic/differential equations. No non-D-finiteness or natural-boundary claim is proved here. [Added 29 September 2026, batch 43: the non-D-finiteness half is answered in Part II. Neither $\sum \theta(n)z^n$ nor $\sum \theta(n)z^n/n!$ is D-finite, and no arithmetic section of θ is P-recursive (Theorem 16.2); the ordinary generating function is transcendental (Corollary 22.1); and a continuous nonconstant log-periodic profile, such as P , excludes P-recursiveness by itself (Proposition 23.1). Still open: whether $\sum \theta(n)z^n$ has a natural boundary, and if not, how it continues (Question 26.1), and nonlinear polynomial differential equations (Question 26.2).]

[Added 29 September 2026, batch 43: Part II proposes nine further questions (Section 26). Three of them approach items of this list from Part II’s side: Question 26.3 (items 1 and 2), Question 26.4 (item 5) and Question 26.7 (item 7).]

13 Conclusion

A nonexistent growth constant need not signal unstructured behavior. For 3AP-free permutations, the balanced combinatorial recurrence yields a unique continuous logarithmic phase profile with a bounded correction. Published finite data do more than distinguish two limits: they certify the profile’s exact fundamental period. That finite certificate determines the entire real scaling-symmetry group and forces exponential gain and loss at every off-dyadic scale.

The explicit 2:1 gluing inequalities show a concrete consequence for combinatorial constructions, while the empirical laws explain which forms of averaging preserve or erase the cutoff phase. These results leave the fine regularity, precise extrema, bounded corrections, and monotonicity of the counting sequence as well-defined further problems rather than concealing them inside an unsupported asymptotic ansatz.

14 Provenance and place in the collection [write]

14.1 Source

[Updated 29 September 2026, batch 43: this section describes Part I. The report now has two parts, built from two manuscripts; Part II’s source, the changes made to it and the choices of that merge are in Section 15.] This report was built from **one** manuscript: manuscript 06 of the fortieth intake batch of PROVEIT’s incoming reports, delivered as the archive `ProveIt_Dyadic_Scaling_Rigidity.zip` (arrival commit 017e6c0d8), dated September 28, 2026, and pinned to PROVEIT revision 74f7f5bdb (full hash in Section 1.3 and in the bibliography). It was placed as a new single-source report in commit `afb2d1227`. It contributed everything in this article except the text marked [write]. No merge was involved, so there were no choices between sources.

When the report was written, every label received the prefix `dsr::`; the text marked [write] was added (the collection note, the correction in Remark 1.1, the notation table of Section 1.5, Remarks 2.4 and 9.2, Section 11, short notes in Sections 1, 10 and 12 and in the appendix, and this section). No statement, proof, number, limitation or non-claim of the manuscript was removed or changed, and no symbol was renamed. The delivered program, data, figure and audit are shipped byte-identical; their shipped paths are listed in the report’s README, and the delivered audit `SOURCE_AND_PROOF_AUDIT.md` still says that the Lean catalogue was not checked for proofs, which Remark 1.1 corrects. The delivered PDF and checksum ledger are not shipped; this article’s PDF is a build of this text.

14.2 Neighbouring reports

No other report in PROVEIT’s collections treats A003407 or 3AP-free permutations. Within `oeis-sequence-asymptotics`, the closest contrast is `a158415-growth-constant`, where a growth constant $\lim a(n)^{1/n}$ is proved to exist; here none exists, and the periodic profile P takes its place. The formal neighbour is the Lean development under `Combinatorics/Ramsey/Lean/` described in Section 11.

Part II. Non-holonomicity of the counts

Sections 1–14 and Appendix A form the original report (Part I) of 28 September 2026. Part I is unchanged except for dated pointers to this addition (in the collection note, after Remark 9.2, in Section 12 and in Section 14), a new Subsection 11.4 of its formal-status Section 11, a line in the title, contents entries for the two parts, Part II’s macros in the preamble, four bibliography entries used only by Part II and two notes in Part I’s entries for Ho and OEIS. No number of Part I moves. Part II was added on 29 September 2026 in batch 43 of PROVEIT’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the non-D-finiteness half of Part I’s research item 8 (Section 12), which ended “No non-D-finiteness or natural-boundary claim is proved here”: $\theta(n)$ is not P-recursive, nor is any arithmetic section $\theta(qn + r)$, and neither generating function is D-finite. The natural-boundary half remains open. Section 15 records the provenance and fixes the notation of the addition; Sections 16–27 and Appendices B–C are the manuscript.

Non-holonomicity of 3AP-Free Permutation Counts

Finite-window rigidity, arithmetic sections,
and stable obstructions to polynomial recurrences

Prepared for Vladimir Reshetnikov · 29 September 2026

Main conclusion. Let $\theta(n)$ count permutations of $[n]$ containing no three-term arithmetic progression as a subsequence. Then $\theta(n)$ is not P-recursive over $\mathbb{C}$. Neither is $\theta(qn + r)$ for any integers $q \geq 1, r \geq 0$. The obstruction survives positive polynomial combinations of finitely many shifted counts and every subexponential multiplicative perturbation with suitable arithmetic coefficients.

Proof status. Conventional mathematical proofs using published G-function theorems and published enumerative inputs. AI-assisted research draft; not refereed and not formalized in Lean or Rocq. Exact verification code and its recorded output accompany the article. The new application was not found in the sources inspected; exhaustive priority is not certified.

Abstract of Part II. We establish non-P-recursiveity of the number $\theta(n)$ of three-term-arithmetic-progression-free permutations, OEIS A003407, and of every arithmetic section $\theta(qn + r)$. This answers the non-D-finiteness question explicitly left outside the conclusions of PROVEIT’s recent report on dyadic scaling rigidity. The missing implication is not simply that oscillatory exponential growth contradicts a polynomial recurrence: even positive integer sequences with rational generating functions can have nonconvergent n th roots. We instead prove a finite-window rigidity lemma. A nonzero leading Nilsson expansion has a fixed-length window whose maximum is bounded above and below by one exponential scale times a power-logarithmic factor. Consequently, a positive, exponentially bounded, integer P-recursive sequence with subexponential forward increases must have an exponential growth rate. For θ , deleting the largest entry gives the required one-sided local estimate, whereas Boon Suan Ho’s 2026 theorem supplies nonconvergence. We reproduce its dyadic separation from two published counts and verify the comparison by exact integers. We then prove persistence under arithmetic sampling, positive polynomial observables, and multiplicative errors $\exp(o(n))$. Both ordinary and exponential generating functions are non-D-finite. We distinguish these results from natural-boundary and differential-transcendence questions, give an independently checked small-count computation, and propose a focused research programme.

15 Provenance and conventions of Part II [write]

15.1 One manuscript, one addition

Part II is batch-43 manuscript 06, *Non-holonomicity of 3AP-Free Permutation Counts*, subtitled *Finite-window rigidity, arithmetic sections, and stable obstructions to polynomial recurrences*, a research article “prepared for Vladimir Reshetnikov” and dated 29 September 2026 (21 A4 pages as delivered); its delivery README calls it AI-assisted and unrefereed. It arrived as the archive `ProveIt_3AP_Nonholonomicity.zip` (inner directory of the same name, main file `article.tex`) in PROVEIT commit `acf39bd6d`, the second arrival commit of batch 43, and was placed in `faef2ed2a`. It was written against PROVEIT commit `8d936ee2357f9decf78c2ecc7d9baf3100a6787d`, reading Part I’s `article.tex` and `README.md` at that commit. Both files are unchanged between the pin and the placement commit (the same Git blobs), so every statement the manuscript makes about “the prior report” or “the repository report” refers to Part I as printed above, before the dated pointers were added.

The manuscript’s code, data and audit are shipped beside Part I’s files with the prefix `02-nonholonomicity-` (Part I’s unprefixes files count as 01): its verifier and unit tests as `code/02-nonholonomicity-verify.py` and `code/02-nonholonomicity-test_verify.py`, its package-root build script as `code/02-nonholonomicity-build.sh`, its recorded outputs as `data/02-nonholonomicity-verification.json` and `data/02-nonholonomicity-unit-tests.txt`, and its source and proof audit as `02-nonholonomicity-SOURCE_AND_PROOF_AUDIT.md`, all byte-identical to the delivery. The manuscript, its delivery README and its PDF are not shipped; they survive in the arrival commit. The package had no checksum ledger. The report’s README maps the delivery names to the shipped paths and gives rerun instructions that do not touch Part I’s files.

The manuscript contributes everything in Sections 16–27 and Appendices B–C, apart from text marked [write], and the title block and abstract printed at the head of this part: the finite-window rigidity lemma and local-growth rigidity theorem (Section 19); non-P-recursiveness of θ and of every arithmetic section (Section 20); the robust, rounded-power, polynomial-observable and Nilsson-layer extensions (Section 21); non-D-finiteness of both generating functions and transcendence of the ordinary one, with their limits (Section 22); the abstract profile obstruction and perturbed linear sampling (Section 23); an exact verification suite with two finite recurrence-exclusion determinants (Section 24); a dependency audit and formalization plan (Section 25); and nine research questions (Section 26).

15.2 What was changed in the manuscript’s text

Every one of the manuscript’s 65 labels received the prefix `dsr:nh:`, and every `\ref` and `\eqref` was updated. Its title page and abstract are printed above as the front matter of this part; its table of contents is merged into the report’s. Its sections and theorems are renumbered by their position (its Section 1 is Section 16 here, its Theorem 1.2 is Theorem 16.2), and its Appendices A and B are Appendices B and C. Its bibliography is merged into the report’s: its entries for Davis et al., Sharma, Correll–Ho, Ho, Hwang–Janson–Tsai and OEIS A003407 are the same works as Part I’s entries and are cited under Part I’s numbers, with the details only the manuscript gives (the journal DOI of Ho’s paper and its own consultation date of the OEIS entry) added to those entries; the entries for Stanley, Garoufalidis (two), and André are new. The manuscript’s entry for the prior report, the present Part I (“PROVEIT research-report collection, *Dyadic scaling rigidity and log-periodic growth of 3AP-free permutations*, report dated 28 September 2026, article and README”), is printed as a bracketed reference to Part I

with the section or theorem that the sentence means; that entry’s caveat stays in force: Part I is unrefereed and unformalized, and no result unique to it is needed for Part II’s main proof. The macro for the pinned hash is expanded, and the manuscript’s colour title page is set in the report’s style. Labels were added to its conclusion, its second appendix and five of its research questions, so that Part I can point to them. The notes marked [write] in Part II cross-reference Part I and the Lean development. No statement, proof, number, limitation or non-claim of the manuscript was removed or changed, and no symbol was renamed.

15.3 Notation across the two parts

Part II keeps the manuscript’s symbols. Most agree with Part I: θ (Part II also writes it θ , the same symbol), $a_n = \log \theta(n)$, $L = \log 2$, $U = \log 21$, $D = \log(21/2)$, $\gamma = \log_2(21/2)$, the piecewise-affine interpolation $A(x)$, the profile P , the bounded correction Q and the subset-state count $C(S)$ mean the same thing in both parts, with the same normalization: Part II’s (23.3) is Part I’s (3.4) at integers. The following letters differ.

Symbol	Meaning in Part II	In Part I; tempting false reading
τ_n	the toll $a_n - a_{\lfloor n/2 \rfloor} - a_{\lceil n/2 \rceil}$, (17.5)	Part I writes this toll r_n (2.4); Part I’s τ (proof of Theorem 5.2) is a hypothetical irrational period
r, q	residue and step of the arithmetic section $\theta(qn + r)$, integers with $q \geq 1$, $r \geq 0$ (r need not be $< q$); in Corollary 23.2, $q > 0$ is real	r is not Part I’s toll r_n ; Part I’s q is a period denominator (Theorem 5.2) or a geometric ratio (Theorem 8.2)
$T(x)$	interpolated toll (Section 23)	Part I’s $R(x)$ (3.2); not the published counts $T_{64} = \theta(64)$, $T_{75} = \theta(75)$ of (18.3)
$H(x), H_n$	$H(x) = \lim 2^{-k} A(2^k x)$ (23.2); in the proof of Lemma 19.3, $H_n = R^n n^\alpha (\log n)^\beta$	$H(x)$ is Part I’s $F(x)$ (3.7)
$F(z), E(z)$	ordinary and exponential generating functions (16.3)	not Part I’s profile limit $F(x)$, nor its gain and loss sets $E_c^\pm$ (Theorem 6.2)
R	exponential scale of a hypothetical Nilsson layer (19.1), “not a radius”	not Part I’s interpolated toll $R(x)$ nor its certificate ratios $R_\pm$
Φ	a polynomial in $\mathbb{Z}_{\geq 0}[N, X_0, \dots, X_s]$ (Theorem 16.4)	Part I’s $\Phi(s) = P(\log_2 s)$ is the phase function on $[1, 2]$
b_n	a local name with four meanings: a least common denominator (Section 19), $\log \theta(qn + r)$ (proof of Theorem 16.3), $\theta(qn + r)$ (proof of Theorem 16.4), $\theta(n)/n!$ (Section 22)	Part I’s $b_n = a_n/n$ throughout
$\Delta_n; d$	$\Delta_n = a_{n+1} - a_n$; d is a recurrence order (16.2), a real exponent (Theorem 16.3), a common difference (24.1), or an X -degree	Part I writes Δ_n as d_n (9.1)
ℓ_m	$\lim_t a_{m2^t} / (m2^t)$, equal to $P(\log_2 m)$	an exponential rate, not a Lean name
s	number of dominant frequencies (Section 19); elsewhere a fixed shift bound	Part I’s $s \in [1, 2]$ is a phase-ray parameter (4.3)
$M_{n,(j,k)}$	row entries $n^k \theta(n + j)$ of a recurrence-exclusion matrix (Section 24)	not Part I’s Cesàro limit $M(x)$, nor the Lean count $\mathbf{Mn}$
$\mathcal{A}(S)$	admissible next values (24.1)	Part I’s test (10.1), written there for a value y ; not the interpolation $A(x)$

The terms “P-recursive” (with an arbitrary starting index), “D-finite” and “Nilsson layer” are Part II’s only; they are defined in Section 16 and (19.1).

15.4 Relation to Part I

Part II uses Part I's objects and, as its own text says, re-derives some of Part I's material as inputs; the credit is Part I's (and, before it, the literature's):

- the deletion bound of Lemma 17.1 is the last assertion of Proposition 9.1, with the same injection;
- the splitting estimates of Proposition 17.2 are Part I's (2.1) and (2.2), the parity construction being the same argument, and the exponential bounds of Lemma 17.3 are Part I's (2.6);
- the binary-descent Lemma 17.4 is Proposition 9.1, plus the fixed-shift consequence (17.8);
- the profile of Section 23 is Theorem 3.2, reproved by the same interpolation, and $\ell_m = P(\log_2 m)$ is the exact-dyadic-ray case of Theorem 4.1;
- the subset-state recurrence of Proposition 24.1 is Part I's (10.2), with a separately written proof and program.

Ho's nonconvergence (Theorem 18.2) is reproduced from $\theta(64)$ and $\theta(75)$; within this report it also follows from Part I's Theorems 4.1 and 5.2, whose certificate uses $\theta(128)$ and $\theta(n)$ for $152 \leq n \leq 182$. Both routes rest on published counts that PROVEIT does not certify (Subsection 11.4). The new content is the passage from nonconvergence plus the deletion bound to non-P-recursiveness, and its extensions.

15.5 Where the merge had to choose

Part II is printed after Part I's closing Section 14 and before Part I's appendix, so no section, theorem or equation number of Part I moves; the manuscript's appendices follow Part I's as Appendices B and C. The formal status of Part II is recorded in Part I's formal-status section, as the new Subsection 11.4, so that one section of the report says what PROVEIT has and has not proved; the manuscript's own dependency audit (Section 25) is kept and points there. Where the manuscript and Part I print the same fact (Subsection 15.4), both statements and both proofs are kept, since the manuscript's versions are the inputs its proofs cite. Part I's question on generating functions is re-scoped, not deleted: its non-D-finiteness half is answered, and its natural-boundary half is Question 26.1 of Part II.

16 The question and the contribution

A permutation $\pi = (\pi_1, \dots, \pi_n)$ of $[n] = \{1, \dots, n\}$ is *3AP-free* if no indices $i < j < k$ satisfy

$$\pi_i + \pi_k = 2\pi_j. \quad (16.1)$$

The positions need not be consecutive, and progressions in both numerical directions are forbidden. Let $\theta(n)$ count these permutations, with $\theta(0) = 1$. The first values are

$$1, 1, 2, 4, 10, 20, 48, 104, 282, 496, 1066, 2460, 6128, \dots$$

This is OEIS A003407 [7].

The distinction between understanding a sequence's growth and understanding its exact generating function is particularly sharp here. Ho proved that $\theta(n)^{1/n}$ has no limit [5]. The recent PROVEIT report *Dyadic Scaling Rigidity and Log-Periodic Growth of 3AP-Free Permutations* develops a continuous periodic growth profile and its consequences, but explicitly makes no non-D-finiteness or natural-boundary claim [Part I, Section 12, item 8]. We resolve the former issue, not the latter.

16.1 Precise statements

Definition 16.1. A sequence (u_n) is *P-recursive over a field K* if there are polynomials $p_0, \dots, p_d \in K[n]$, not all zero, such that

$$\sum_{j=0}^d p_j(n) u_{n+j} = 0 \quad (16.2)$$

for every sufficiently large integer n . A formal power series is *D-finite over $K(z)$* if it satisfies a nonzero homogeneous linear differential equation with coefficients in $K[z]$.

Allowing an arbitrary starting index in (16.2) is important: non-P-recursive below excludes eventual recurrences, not merely recurrences valid from $n = 0$.

Theorem 16.2 (Main result). *For every pair of integers $q \geq 1$ and $r \geq 0$, the sequence*

$$(\theta(qn + r))_{n \geq 0}$$

is not P-recursive over $\mathbb{C}$. In particular, neither

$$F(z) = \sum_{n \geq 0} \theta(n) z^n, \quad E(z) = \sum_{n \geq 0} \theta(n) \frac{z^n}{n!} \quad (16.3)$$

is D-finite over $\mathbb{C}(z)$.

The ordinary generating function has positive finite radius of convergence; the exponential generating function is entire. Thus their non-D-finiteness concerns two rather different analytic regimes.

Theorem 16.3 (Robustness under approximation). *Fix integers $q \geq 1$, $r \geq 0$ and a real number $d > 0$. There is no eventually positive, integer-valued P-recursive sequence (v_n) for which*

$$\log v_n = d \log \theta(qn + r) + o(n). \quad (16.4)$$

The same conclusion holds for positive rational v_n if the least common denominator of $v_0, \dots, v_n$ is at most C^{n+1} for some fixed C .

The approximation condition is pointwise as $n \rightarrow \infty$, not an average condition or a condition on selected subsequences. It rules out, for example, a P-recursive positive integer surrogate within a polynomial factor of the counts. It does not rule out approximations with unrestricted real coefficients or uncontrolled rational denominators.

Theorem 16.4 (Positive polynomial observables). *Let $q \geq 1$, $r, s \geq 0$ be integers, and let*

$$\Phi \in \mathbb{Z}_{\geq 0}[N, X_0, \dots, X_s]$$

be a polynomial that depends on at least one X_j . Then

$$v_n = \Phi(n, \theta(qn + r), \theta(qn + r + 1), \dots, \theta(qn + r + s)) \quad (16.5)$$

is not P-recursive over $\mathbb{C}$.

This includes powers of the counts, products of finitely many shifted counts, and positive polynomially weighted finite sums. Nonnegative coefficients are a genuine hypothesis: the argument does not control cancellation in signed combinations.

16.2 What is new here, and what is imported

The intended contribution is the application of finite-window asymptotic rigidity to A003407, including its sections and stable transformations. The nonexistence of a growth constant is Ho’s theorem [5]; it is not a new result of this article. The periodic-profile construction belongs to existing divide-and-conquer theory [6] and the prior repository report [Part I, Theorem 3.2]. The arithmetic regularity and Nilsson asymptotics of G-functions are established theory [10, 11, 12]. We use them as explicit inputs, not as conjectures or newly proved foundations.

A targeted search for A003407 and 3AP-free permutations together with “P-recursive”, “holonomic”, and “D-finite”, and inspection of the papers cited here, did not locate the present application. Some searches returned unrelated material, so this is not evidence of exhaustive coverage. The claim of progress is precise: the stated repository gap is filled by a complete argument from identified published results. Priority beyond that assessment remains subject to specialist checking.

17 The enumerative inputs

We record the ingredients independently of the later analytic argument. Unless a base is displayed, logarithms are natural.

17.1 A local estimate needing no structural classification

Lemma 17.1 (Deletion bound). *For every $n \geq 0$,*

$$\theta(n+1) \leq (n+1)\theta(n). \quad (17.1)$$

Consequently, for fixed $j \geq 0$ and $n \geq 1$,

$$\theta(n+j) \leq (n+j)^j \theta(n). \quad (17.2)$$

Proof. Delete the largest entry $n+1$ from a 3AP-free permutation of $[n+1]$. The remaining permutation of $[n]$ is still 3AP-free, because deleting an entry cannot create a previously absent subsequence. The original permutation is determined by the remaining permutation and the position at which $n+1$ was inserted. There are at most $n+1$ positions. This is an injection into a set of cardinality $(n+1)\theta(n)$. Iterating the estimate proves (17.2). $\square$

No monotonicity assertion is used. In particular, we do not assume $\theta(n+1) \geq \theta(n)$.

[write] This is the deletion bound at the end of Part I’s Proposition 9.1. PROVEIT proves the stronger bound $\theta(n+1) \leq \lfloor (n+3)/2 \rfloor \theta(n)$, $n \geq 1$, in Lean as `insertion_count_upper_bound_holds` (Subsection 11.4); the argument below needs only the weaker one.

17.2 The balanced splitting inequalities

Proposition 17.2 (Classical splitting estimates). *For every integer $n \geq 2$,*

$$2\theta(\lfloor n/2 \rfloor)\theta(\lceil n/2 \rceil) \leq \theta(n) \leq 21\theta(\lfloor n/2 \rfloor)\theta(\lceil n/2 \rceil). \quad (17.3)$$

The lower estimate is the parity construction of Davis, Entinger, Graham, and Simmons [1]. For completeness, order the even entries and the odd entries separately by affine copies of smaller 3AP-free permutations, and concatenate the blocks in either order. The endpoints of an integer three-term progression have the same parity. They either lie in a common 3AP-free

block with their midpoint, or their midpoint lies in the other block and cannot occur between them. The two block orders give disjoint families, proving the factor 2.

The upper estimate for $n \geq 3$ is Sharma's Theorem 2.8 [2]; its proof is imported. At $n = 2$ it follows from $\theta(2) = 2$. We do not infer this upper estimate from a fit to finite data.

[write] Proposition 17.2 is Part I's (2.1) and (2.2). In PROVEIT both halves are Lean theorems: the upper one for $n \geq 3$ is `LeanProofs.Sharma2012.theorem_2_8_holds`, the lower one is `even_count_recurrence_holds` and `odd_count_recurrence_holds` (Remark 2.4). Nothing derived from them below is formalized (Subsection 11.4).

Write

$$a_n = \log \theta(n), \quad L = \log 2, \quad U = \log 21, \quad D = U - L, \quad (17.4)$$

and, for $n \geq 2$, define

$$\tau_n = a_n - a_{\lfloor n/2 \rfloor} - a_{\lceil n/2 \rceil}. \quad (17.5)$$

Thus $L \leq \tau_n \leq U$.

Lemma 17.3 (Exponential bounds). *For $n \geq 1$,*

$$2^{n-1} \leq \theta(n) \leq 21^{n-1}. \quad (17.6)$$

Proof. The logarithmic version holds at $n = 1$. In a strong induction step, the two smaller indices sum to n , so their lower bounds sum to $(n-2)L$. Adding $\tau_n \geq L$ gives $(n-1)L$. The upper bound is identical with U in place of L . $\square$

The much sharper published upper bound $\theta(n) \leq 2.7^n/21$ for $n \geq 11$ [2] is unnecessary for our argument. The deliberately weak bound (17.6) already provides all the arithmetic growth control needed below.

17.3 A two-sided shift estimate for the stronger corollaries

The main non-P-recursiveness theorem needs only Lemma 17.1. Positive polynomial observables will also use a lower comparison for nearby counts.

Lemma 17.4 (Binary descent for adjacent logarithms). *Put $\Delta_n = a_{n+1} - a_n$ for $n \geq 1$, and $\gamma = D/\log 2 = \log_2(21/2)$. Then*

$$|\Delta_n - \log 2| \leq D \lfloor \log_2 n \rfloor, \quad 2n^{-\gamma} \leq \frac{\theta(n+1)}{\theta(n)} \leq 2n^\gamma. \quad (17.7)$$

For each fixed integer $s \geq 0$,

$$\max_{0 \leq j \leq s} |a_{n+j} - a_n| = O_s(\log(n+1)). \quad (17.8)$$

Proof. For $m \geq 1$, subtracting two consecutive instances of (17.5) gives

$$\Delta_{2m} = \Delta_m + \tau_{2m+1} - \tau_{2m}, \quad (17.9)$$

$$\Delta_{2m+1} = \Delta_m + \tau_{2m+2} - \tau_{2m+1}. \quad (17.10)$$

Thus $\Delta_n = \Delta_{\lfloor n/2 \rfloor} + \varepsilon_n$ for $n \geq 2$, where $|\varepsilon_n| \leq D$. Iterating reaches $\Delta_1 = \log 2$ after $\lfloor \log_2 n \rfloor$ steps. Exponentiation and summation over a fixed number of adjacent differences give the remaining statements. $\square$

The same descent identity appears in the prior repository report [Part I, Proposition 9.1]; it is restated and reproved here as an input, not claimed as a new result.

18 The two distinct exponential scales

The only numerical facts used in the infinite nonconvergence argument are the two published counts displayed below. The deduction from them is exact.

Lemma 18.1 (Dyadic limits and their enclosures). *For every $m \geq 1$, the sequence*

$$\frac{a_{m2^t}}{m2^t} \quad (t \geq 0)$$

converges to a finite number ℓ_m , and

$$\frac{\log(2\theta(m))}{m} \leq \ell_m \leq \frac{\log(21\theta(m))}{m}. \quad (18.1)$$

Proof. At even indices the recurrence is $a_{2k} = 2a_k + \tau_{2k}$, so

$$\frac{a_{m2^{t+1}}}{m2^{t+1}} - \frac{a_{m2^t}}{m2^t} = \frac{\tau_{m2^{t+1}}}{m2^{t+1}}.$$

The increments form an absolutely summable series. Summing their lower and upper bounds gives respectively L/m and U/m , proving (18.1). $\square$

The published values used by Ho [5], also in the Correll–Ho/OEIS data [4, 7], are

$$T_{64} := \theta(64) = 39911512393313043466768, \quad (18.2)$$

$$T_{75} := \theta(75) = 30235147387260979648843264. \quad (18.3)$$

They yield the exact strict comparison

$$(2T_{64})^{75} > (21T_{75})^{64}. \quad (18.4)$$

Equivalently,

$$\ell_{64} \geq \frac{\log(2T_{64})}{64} > \frac{\log(21T_{75})}{75} \geq \ell_{75}. \quad (18.5)$$

For a short rational certificate, the accompanying program verifies

$$(2T_{64})1000^{64} > 2279^{64}, \quad (18.6)$$

$$(21T_{75})500^{75} < 1139^{75}. \quad (18.7)$$

Hence $e^{\ell_{64}} > 2.279$ and $e^{\ell_{75}} < 2.278$. In particular,

$$\ell_{64} - \ell_{75} > \log \frac{2279}{2278} > \frac{1}{2279}. \quad (18.8)$$

The last strict inequality follows by integrating $1/t$ on $[1, 1 + 1/2278]$, or from $\log(1+x) > x/(1+x)$ for $x > 0$.

Theorem 18.2 (Ho’s growth nonconvergence). *The sequence $\theta(n)^{1/n}$ has no limit.*

Proof. The subsequences $n = 64 \cdot 2^t$ and $n = 75 \cdot 2^t$ converge to the distinct limits $e^{\ell_{64}}$ and $e^{\ell_{75}}$ by Lemma 18.1 and (18.5). $\square$

This is a reproduction of the published mechanism [5], not a new claim of nonconvergence. The decimal separation between the two enclosure endpoints is approximately 0.00109603287135 on the logarithmic scale. The exact inequalities, not that decimal, are the proof certificates.

Remark 18.3 (Enumeration boundary). The present package independently enumerates $\theta(n)$ only for $0 \leq n \leq 32$. It does not re-enumerate (18.3). Thus the displayed numerical route imports those two established counts. Alternatively, one may take Theorem 18.2 directly from the published paper. These are two presentations of the same dependency, not two independent verifications of the large enumeration.

[write] Nothing in PROVEIT certifies $\theta(64)$ or $\theta(75)$: the Lean count certificate stops at $\theta(20)$ (Subsection 11.4). The two values equal rows 64 and 75 of Part I's published table `data/counts.txt`, and the displayed comparison and brackets were rechecked from those rows when Part II was written; that is the same published data, not a second source. Part I reaches $p_- < p_+$, and hence Theorem 18.2, from other published counts (Theorem 5.2).

19 From a polynomial recurrence to a finite-window law

This section contains the central inference. We separate the published asymptotic input from the finite-dimensional argument that uses it.

19.1 Descending complex recurrence coefficients

Lemma 19.1 (Rational descent). *A rational-valued sequence that is P -recursive over $\mathbb{C}$ is P -recursive over $\mathbb{Q}$.*

Proof. Fix the order d , a coefficient degree bound e , and a starting index N for a complex recurrence. Its coefficients form a nonzero vector in $\mathbb{C}^{(d+1)(e+1)}$ annihilated by every rational row

$$(n^k u_{n+j})_{0 \leq j \leq d, 0 \leq k \leq e}, \quad n \geq N.$$

The rational span of these rows has a basis consisting of finitely many rows. Their matrix has the same rank over $\mathbb{Q}$ and $\mathbb{C}$, since rank is characterized by nonzero minors. Its complex kernel is nonzero, so its rational kernel is nonzero. A nonzero rational vector in that kernel annihilates every original row, producing the required rational recurrence. $\square$

This also makes clear why no restriction to rational recurrence coefficients is hidden in Theorem 16.2.

19.2 The arithmetic asymptotic input

For a rational sequence (u_n) , let b_n be the least positive integer such that $b_n u_0, \dots, b_n u_n$ are all integers. We say that its *common denominators grow exponentially* if $b_n \leq C^{n+1}$ for some C . For integer sequences this holds with $b_n = 1$.

Theorem 19.2 (Published G-function input). *Suppose (u_n) is rational-valued, not eventually zero, and P -recursive over $\mathbb{Q}$, with $|u_n| \leq C^{n+1}$ and exponentially growing common denominators. Then its generating series is a G-function. Its coefficients have a nonzero leading asymptotic layer of the form*

$$u_n = R^n n^\alpha (\log n)^\beta \left(\sum_{j=1}^s c_j \zeta_j^n + o(1) \right), \quad (19.1)$$

where $R > 0$, $\alpha \in \mathbb{Q}$, $\beta \in \mathbb{Z}_{\geq 0}$, the ζ_j are distinct complex numbers of modulus 1, and the c_j are nonzero complex constants.

The G-function criterion and coefficient expansion are the relevant parts of Garoufalidis's Theorem 3 and Proposition 2.5 [10]; the precise nonzero leading-layer convention is explained in [11]. The underlying arithmetic theorem makes the minimal differential operator regular singular with rational exponents [12]. These are the deep imported results. We do not appeal to a conjectural geometric origin of G-functions.

To clarify the reduction to (19.1), start with the finite power-logarithmic families at the dominant singularities, collect equal frequencies, and choose the largest power of n having a nonzero coefficient, then the largest logarithmic power within it. The remaining terms are $o(1)$ after the displayed normalization. Contributions from singularities farther away are exponentially smaller. Regular singularity is essential to this explanation: quasi-unipotent monodromy alone, without control of singular growth, would not justify such a local form.

19.3 Why cancellation cannot persist through a whole window

Lemma 19.3 (Finite-window noncancellation). *Suppose a complex sequence satisfies (19.1). There are constants $c, C > 0$ and n_0 such that, for $n \geq n_0$,*

$$cR^n n^\alpha (\log n)^\beta \leq \max_{0 \leq k < s} |u_{n+k}| \leq CR^n n^\alpha (\log n)^\beta. \quad (19.2)$$

In particular,

$$\lim_{n \rightarrow \infty} \left(\max_{0 \leq k < s} |u_{n+k}| \right)^{1/n} = R. \quad (19.3)$$

Proof. Write $H_n = R^n n^\alpha (\log n)^\beta$. For each fixed k ,

$$\frac{H_{n+k}}{H_n} = R^k (1 + o(1)).$$

Since there are only s values of k , the errors are uniform for $0 \leq k < s$. Define the $s \times s$ matrix

$$V_{kj} = (R\zeta_j)^k, \quad 0 \leq k < s, \quad 1 \leq j \leq s,$$

and the column vector $w_n = (c_j \zeta_j^n)_{j=1}^s$. Then

$$\left(\frac{u_{n+k}}{H_n} \right)_{k=0}^{s-1} = V w_n + o(1) \quad (19.4)$$

in the maximum norm. The Vandermonde determinant is

$$\det V = \prod_{1 \leq i < j \leq s} (R\zeta_j - R\zeta_i) \neq 0.$$

Moreover, $\|w_n\|_\infty = c_* := \max_j |c_j| > 0$ for every n . Thus

$$\|V w_n\|_\infty \geq \frac{c_*}{\|V^{-1}\|_\infty} > 0, \quad \|V w_n\|_\infty \leq \|V\|_\infty c_*.$$

After increasing n_0 , the error in (19.4) is less than half of the lower constant. This proves (19.2). Finally, $H_n^{1/n} \rightarrow R$, because $\log n/n$ and $\log \log n/n$ tend to zero. $\square$

The phases may be roots of unity, algebraic points with irrational arguments, or arbitrary distinct unit complex numbers in this lemma. No equidistribution or Diophantine lower bound for a single oscillating sum is required.

19.4 One-sided local growth is enough

Definition 19.4. An eventually positive sequence has *subexponential forward increases* if

$$\log^+ \frac{u_{n+1}}{u_n} = o(n), \quad \log^+ x = \max\{0, \log x\}. \quad (19.5)$$

This is a one-sided condition. It permits downward jumps, and is weaker than $\log(u_{n+1}/u_n) = o(n)$.

Theorem 19.5 (Local-growth rigidity). *Let (u_n) be eventually positive, rational-valued, exponentially bounded, and have exponentially growing common denominators. If it is P-recursive over $\mathbb{C}$ and satisfies (19.5), then*

$$\lim_{n \rightarrow \infty} u_n^{1/n} = R \quad (19.6)$$

exists in $(0, \infty)$. In particular, the assertion applies to positive integer sequences with $u_{n+1}/u_n \leq Cn^A$ for fixed $C, A > 0$.

Proof. Rational descent and Theorem 19.2 give a leading layer (19.1). Set $W_n = \max_{0 \leq k < s} u_{n+k}$. If $h_n = \log^+(u_{n+1}/u_n)$, then for every $k < s$,

$$\log u_{n+k} \leq \log u_n + \sum_{i=0}^{k-1} h_{n+i}.$$

The finite sum is $o(n)$, uniformly over these k . Since $u_n \leq W_n$,

$$0 \leq \log W_n - \log u_n \leq o(n).$$

Lemma 19.3 gives $\log W_n/n \rightarrow \log R$, so $\log u_n/n \rightarrow \log R$ as well. Exponentiation proves (19.6). $\square$

Corollary 19.6 (A necessary jump phenomenon). *If an eventually positive, exponentially bounded, integer P-recursive sequence has no n th-root limit, then there is $\varepsilon > 0$ such that*

$$\frac{u_{n+1}}{u_n} > e^{\varepsilon n}$$

for infinitely many n .

Proof. Otherwise the nonnegative sequence in the left side of (19.5), divided by n , tends to zero, contradicting Theorem 19.5. $\square$

This is the appropriate obstruction: persistent exponential-scale oscillation in an arithmetic P-recursive sequence requires exponentially large upward transitions somewhere. The deletion estimate prevents them for the permutation counts.

20 Proof of non-P-recursiveness and all arithmetic sections

20.1 The base sequence

Suppose $\theta(n)$ were P-recursive over $\mathbb{C}$. It is positive and integral, and Lemma 17.3 bounds it exponentially. Furthermore, (17.1) implies

$$\log^+ \frac{\theta(n+1)}{\theta(n)} \leq \log(n+1) = o(n).$$

Theorem 19.5 would force $\theta(n)^{1/n}$ to converge, contradicting Theorem 18.2. This proves the base case of Theorem 16.2.

20.2 A bounded-gap interpolation principle

The all-sections assertion is stronger than the base case: a non-P-recursive sequence may have P-recursive sections. Here the local estimate prevents that escape.

Lemma 20.1 (A growth rate on one arithmetic section forces a global rate). *Let (u_m) be positive and exponentially bounded, with subexponential forward increases. Fix integers $q \geq 1$, $r \geq 0$. If*

$$\lim_{n \rightarrow \infty} u_{qn+r}^{1/n} = B \in (0, \infty),$$

then $u_m^{1/m} \rightarrow B^{1/q}$ as $m \rightarrow \infty$.

Proof. For large m , choose n with $x = qn+r \leq m < y = q(n+1)+r$. Let $h_k = \log^+(u_{k+1}/u_k) = o(k)$. At most q consecutive terms occur in either of the following sums, so

$$\log u_m \leq \log u_x + \sum_{k=x}^{m-1} h_k = \log u_x + o(m),$$

and

$$\log u_m \geq \log u_y - \sum_{k=m}^{y-1} h_k = \log u_y - o(m).$$

Now $n/m \rightarrow 1/q$, and the two bounding logarithms, divided by m , both tend to $(\log B)/q$. The conclusion follows by squeezing. $\square$

Proof of the arithmetic-section part of Theorem 16.2. Fix q, r and write $v_n = \theta(qn+r)$. This is a positive integer sequence bounded above by 21^{qn+r} . Repeated deletion yields

$$\frac{v_{n+1}}{v_n} \leq \prod_{j=1}^q (qn+r+j) \leq (qn+r+q)^q.$$

If v_n were P-recursive, Theorem 19.5 would give a finite positive limit for $v_n^{1/n}$. Lemma 20.1 would then give a global n th-root limit for $\theta(n)$, contradicting Theorem 18.2. $\square$

Notice that neither the continuous phase profile nor the two-sided estimate (17.7) is needed for this proof. The stronger all-sections conclusion already follows from one-sided insertion control.

20.3 Explicit separated limits within every section

For later applications it is useful to see the two scales directly. For large t , let

$$n_t^{(m)} = \left\lfloor \frac{m2^t - r}{q} \right\rfloor, \quad m \in \{64, 75\}.$$

The number $qn_t^{(m)} + r$ differs from $m2^t$ by less than q . By Lemma 17.4, their logarithmic counts differ by $O_q(t+1)$. Therefore

$$\frac{\log \theta(qn_t^{(m)} + r)}{n_t^{(m)}} \rightarrow q\ell_m. \quad (20.1)$$

The two limiting logarithmic rates are separated by more than $q/2279$. This construction provides explicit infinite subsequences; it does not rely on identifying the global extrema of the phase profile.

21 Stable non-holonomicity

21.1 Subexponential multiplicative perturbations

Proof of Theorem 16.3. Assume a sequence v_n as in the statement exists, and put $b_n = \log \theta(qn + r)$ and $e_n = \log v_n - db_n$. Then $e_n = o(n)$. The exponential bound for θ shows $v_n \leq C^{n+1}$ after increasing C to absorb a finite prefix. Repeated deletion gives $b_{n+1} - b_n \leq q \log(qn + r + q)$. Hence

$$\log \frac{v_{n+1}}{v_n} = d(b_{n+1} - b_n) + e_{n+1} - e_n \leq O(\log n) + |e_{n+1}| + |e_n| = o(n).$$

Thus v_n has subexponential forward increases. Its assumed integrality, or its rationality and common-denominator bound, lets us apply Theorem 19.5. Consequently $\log v_n/n$ converges. But (16.4), with $d > 0$, then forces b_n/n to converge, contradicting Lemma 20.1 and Theorem 18.2. $\square$

Corollary 21.1. *For every real $d > 0$ and integers $q \geq 1$, $r \geq 0$, the integer sequence*

$$\lfloor \theta(qn + r)^d \rfloor$$

is not P-recursive. In particular, every positive integral power of every arithmetic section is non-P-recursive.

Proof. The lower bound (17.6) makes $\theta(qn + r)^d$ exponentially large. Taking its floor changes its logarithm by $o(1)$; apply Theorem 16.3. $\square$

Rounding is used only to produce integer coefficients. The assertion does not require d to be computable or algebraic, although an arbitrary real d need not furnish an effective algorithm for its rounded sequence.

21.2 Positive polynomial observables

Proof of Theorem 16.4. Let $d \geq 1$ be the largest total degree in the X -variables of a monomial occurring in Φ . Put $b_n = \theta(qn + r)$. For each fixed $j \leq s$, Lemma 17.4 gives

$$\theta(qn + r + j) = b_n \exp(O(\log n)).$$

A monomial $cN^k \prod_{j=0}^s X_j^{m_j}$ with $h = \sum_j m_j$ and $c > 0$ therefore evaluates to

$$b_n^h \exp(O(\log n)).$$

For $n \geq 1$, every factor is positive. Select one occurring monomial of X -degree d . It gives a lower bound $b_n^d n^{-A}$ for a sufficiently large constant A . Since $b_n \geq 1$, every occurring monomial is at most $b_n^d n^A$ after enlarging A ; summing the finitely many monomials gives an upper bound of the same form. Thus

$$\log v_n = d \log b_n + O(\log n). \quad (21.1)$$

The sequence is eventually positive and integer-valued. Apply Theorem 16.3. $\square$

For example,

$$\theta(n)^2 + 7n^3\theta(n+1), \quad \theta(3n+2)\theta(3n+5), \quad \sum_{j=0}^s (n+1)^j \theta(n+j)$$

are all non-P-recursive. In the first expression the term of X -degree 2 determines the exponential scale; in the second, missing intermediate shifts simply mean that their exponents in the polynomial are zero.

21.3 The same obstruction excludes a leading Nilsson description

Corollary 21.2. *For every $q \geq 1$, $r \geq 0$, the sequence $\theta(qn + r)$ cannot have a nonzero leading layer of the form (19.1). The same holds for the sequences in Theorem 16.4.*

Proof. A leading layer alone, without any P-recursive hypothesis, gives the window law by Lemma 19.3. The local-growth part of the proof of Theorem 19.5 would then force an n th-root limit. For sections this contradicts (20.1). For polynomial observables, (21.1) gives two limits $dq\ell_{64}$ and $dq\ell_{75}$, and also gives subexponential forward increases. $\square$

Thus replacing a hoped-for exact recurrence by a finite dominant power-logarithmic asymptotic ansatz does not remove the obstruction.

22 Generating-function consequences and their limits

22.1 Ordinary and exponential generating functions

The standard coefficient–differential-equation equivalence says that $\sum u_n z^n$ is D-finite exactly when (u_n) is P-recursive [9, Theorem 1.5]. Theorem 16.2 therefore proves non-D-finiteness of F and of the ordinary generating function of every section and every positive polynomial observable above.

For the exponential generating function, write $b_n = \theta(n)/n!$. Multiplying a recurrence for b_n by $(n + d)!$ gives

$$\sum_{j=0}^d p_j(n) \frac{(n + d)!}{(n + j)!} \theta(n + j) = 0,$$

whose coefficients are polynomials. Conversely, a recurrence for $\theta(n)$ can be divided by $n!$ to give one for b_n , because $(n + j)!/n!$ is a polynomial in n . These transformations preserve nontriviality. Hence E is D-finite if and only if F is D-finite, which completes the proof of (16.3)’s assertion.

The bounds (17.6) show that F has radius in $[1/21, 1/2]$. They also give absolute convergence of E everywhere, since $|E(z)| \leq \sum 21^n |z|^n / n! = e^{21|z|}$. The latter bound is deliberately nonoptimal.

Corollary 22.1. *The ordinary generating function of every sequence in Theorems 16.2 and 16.4 is transcendental over $\mathbb{C}(z)$.*

Proof. An algebraic germ f belongs to the finite extension $\mathbb{C}(z, f)$ of $\mathbb{C}(z)$. In characteristic zero, differentiation preserves that extension: implicit differentiation of a separable minimal polynomial expresses f' in the same field, and iteration does so for every derivative. Finitely many derivatives must therefore be linearly dependent over $\mathbb{C}(z)$. Thus an algebraic germ is D-finite, contrary to the established results. $\square$

22.2 Why nonconvergence by itself was insufficient

Example 22.2 (A positive rational counterexample to the naive argument). Let

$$u_n = 3^n + (-3)^n + 2^n.$$

For even n , $u_n = 2 \cdot 3^n + 2^n$; for odd n , $u_n = 2^n$. The sequence is positive and integral, its n th roots tend to 3 on even indices and to 2 on odd indices, and yet its generating function is

$$\frac{1}{1 - 3z} + \frac{1}{1 + 3z} + \frac{1}{1 - 2z}.$$

It satisfies a constant-coefficient recurrence. The ratio from an odd index to the next even index is exponentially large, exactly as Corollary 19.6 requires. A two-term window cannot remain small.

The finite-window argument is consequently not a cosmetic addition. It eliminates a real counterexample mechanism that a bare appeal to oscillatory growth would miss.

22.3 No natural-boundary claim

A non-D-finite function need not have a natural boundary. For a direct example, the meromorphic function $1/\cos z$ has isolated poles at $\pi/2 + k\pi$, and it can be continued around each of them. It is not D-finite, since a polynomial-coefficient linear differential equation has only finitely many possible finite singularities, whereas these poles are infinite in number. Thus non-D-finiteness does not say that continuation fails across every point of a convergence circle.

Our result also does not prove differential transcendence: nonlinear polynomial differential equations are not excluded. The radius of F has not been shown transcendental or irrational. Nor does the argument prove that F has infinitely many singularities. What it excludes, in particular, is the finite regular-singular arithmetic configuration that would be forced by a G-function.

23 The periodic profile as a reusable obstruction

The proof of the main theorem was deliberately independent of the prior report's stronger profile results. This section connects the new criterion to that report without treating its established construction as new.

23.1 The abstract profile criterion

Proposition 23.1 (Continuous nonconstant phase obstruction). *Let (u_n) be positive integers and suppose, for some $b > 1$ and a continuous nonconstant 1-periodic function P , that*

$$\log u_n = nP(\log_b n) + o(n). \quad (23.1)$$

Then (u_n) is not P -recursive over $\mathbb{C}$. The same conclusion holds for positive rational coefficients with exponentially growing common denominators.

Proof. Periodicity and continuity make P bounded and uniformly continuous, so (23.1) bounds u_n exponentially. Writing $e_n = o(n)$,

$$\begin{aligned} \log u_{n+1} - \log u_n &= P(\log_b(n+1)) \\ &\quad + n(P(\log_b(n+1)) - P(\log_b n)) + e_{n+1} - e_n = o(n). \end{aligned}$$

Thus subexponential forward increases hold. For any fixed $t \in \mathbb{R}$, take $n_k = \lfloor b^{k+t} \rfloor$. Then $\log_b n_k - k \rightarrow t$, and $\log u_{n_k}/n_k \rightarrow P(t)$. Two distinct values of P give two distinct logarithmic root limits, contradicting Theorem 19.5 under P -recursiveness. $\square$

Only continuity is required, not differentiability, Lipschitz regularity, or identification of the fundamental period. In particular, the unresolved fine regularity questions in the prior report do not obstruct this application.

23.2 Recovering the repository's bounded-error law

Here is a short independent specialization of the existing interpolation method [6] [Part I, Theorem 3.2]. Let $A(x)$ be the affine interpolation of a_n for $x \geq 1$, and put $T(x) = A(x) - 2A(x/2)$ for $x \geq 2$. At even and odd integers respectively,

$$2A(n/2) = 2a_{n/2}, \quad 2A(n/2) = a_{(n-1)/2} + a_{(n+1)/2}.$$

On every unit interval T is affine, because a breakpoint of $A(x/2)$ can only occur at an even integer. Hence T interpolates τ_n and $L \leq T(x) \leq U$.

The series

$$H(x) = A(x) + \sum_{j \geq 1} 2^{-j} T(2^j x), \quad x \geq 1, \quad (23.2)$$

converges uniformly. Equivalently, $H(x) = \lim_{k \rightarrow \infty} 2^{-k} A(2^k x)$, so H is continuous and $H(2x) = 2H(x)$. Defining $P(t) = 2^{-t} H(2^t)$ for $t \geq 0$ and extending periodically gives

$$a_n = nP(\log_2 n) - Q(n), \quad L \leq Q(n) \leq U. \quad (23.3)$$

The dyadic limits satisfy $\ell_m = P(\log_2 m)$, so (18.5) proves that this P is nonconstant. Proposition 23.1 is therefore a second route to the base non-P-recursiveness theorem. The profile construction and the conclusion that it is nonconstant are not additional novelty claims here.

[write] In Part I's notation T is R (3.2), H is F (3.7), and (23.3) is Theorem 3.2 at integers, with the same P and Q . Part I's Lemma 5.1 gives a second proof that P is nonconstant, from other published counts.

23.3 Perturbed linear sampling

Corollary 23.2. *Fix a real $q > 0$, and let m_n be positive integers satisfying*

$$m_n = qn + o\left(\frac{n}{\log(n+1)}\right).$$

Then $(\theta(m_n))$ is not P -recursive over $\mathbb{C}$.

Proof. The two-sided adjacent bound gives

$$|a_{m_n} - a_{\lfloor qn \rfloor}| = O((|m_n - \lfloor qn \rfloor| + 1) \log(n+1)) = o(n).$$

Using (23.3) and continuity,

$$\log \theta(m_n) = n q P(\log_2 n + \log_2 q) + o(n).$$

The function $t \mapsto qP(t + \log_2 q)$ is continuous, periodic, and nonconstant. Apply Proposition 23.1. $\square$

The indices need not themselves satisfy a recurrence, and the statement is an obstruction, not an enumeration algorithm. It shows that the phenomenon is not an artefact of choosing a single residue class.

24 Exact verification and reproducibility

The package contains a standard-library Python program, not a symbolic recurrence guesser masquerading as a proof. Its output separates four roles: independent enumeration, arithmetic separation, finite diagnostic recurrence certificates, and the limitations of each check.

24.1 A subset-state enumeration with a proved transition

For a subset $S \subseteq [n]$, regard S as the set of entries already placed. If $x \notin S$ is placed next, then for every integer $d > 0$ with $x - d, x + d \in [n]$, the two endpoints must either both belong to S or both lie outside S . Define

$$\mathcal{A}(S) = \{x \notin S : \mathbf{1}_S(x - d) = \mathbf{1}_S(x + d) \text{ whenever } x - d, x + d \in [n]\}. \quad (24.1)$$

The continuation recurrence is

$$C([n]) = 1, \quad C(S) = \sum_{x \in \mathcal{A}(S)} C(S \cup \{x\}). \quad (24.2)$$

Proposition 24.1. *The value $C(\emptyset)$ in (24.2) equals $\theta(n)$.*

Proof. In any completed ordering, a forbidden progression occurs exactly when its midpoint is placed with one endpoint earlier and the other endpoint later. Condition (24.1) excludes precisely that event at the step when the midpoint is placed. If all steps obey it, no midpoint can lie between its endpoints, so the completed permutation is 3AP-free. Conversely, every 3AP-free permutation obeys the condition at each step. The condition for future steps depends only on S , not on the internal order of an already valid prefix. Therefore memoization by the subset is sound, and (24.2) counts every valid completion exactly once. $\square$

This subset-state approach is consistent with the dynamic-programming viewpoint in Correll–Ho [4]. The present implementation and its short transition proof are provided for independent reproducibility, not as a new faster enumeration method. [write] Part I’s Section 10 proves the same recurrence (10.2), and its own verifier, a different program, also recomputes $\theta(n)$ for $n \leq 32$.

24.2 Executed checks

The supplied run independently computed all counts for $0 \leq n \leq 32$ and matched the published table. A second implementation directly iterated through all permutations for $0 \leq n \leq 8$ and tested (16.1). The two algorithms agree on their overlap. The program also checked the finite instances of (17.3) through 32 and all integer inequalities in Section 18.

n	$\theta(n)$	n	$\theta(n)$
0	1	8	282
1	1	12	6128
2	2	16	212728
3	4	20	2937136
4	10	24	96547832
5	20	28	2181990978
6	48	30	14765405996
7	104	32	121728075232

All integer assertions remain active under `python -O`; the program uses explicit checks rather than optimizable `assert` statements. Its decimal logarithms are diagnostic only.

24.3 Finite recurrence exclusion certificates

For proposed order d and coefficient degree e , form rows

$$M_{n,(j,k)} = n^k \theta(n+j), \quad 0 \leq j \leq d, \quad 0 \leq k \leq e.$$

A recurrence valid on these rows gives a nonzero vector in the kernel of M . The script uses fraction-free Bareiss elimination to compute exact integer determinants and obtains nonzero determinants for the following square matrices:

Order d	Degree e	First row n	Size	Count indices required
5	3	0	24	0, ..., 28
4	3	8	20	8, ..., 31

The full signed determinant integers are in `data/verification.json`. A nonzero determinant excludes every recurrence in the indicated order-degree rectangle on those rows, because smaller rectangles embed by adding zero coefficients.

These are *finite diagnostics*. They do not rule out a recurrence starting after the tested interval, a larger order, or a larger degree. Theorem 16.2, not matrix fitting, excludes all such possibilities.

24.4 Build and run instructions

From the package root:

```
python3 code/verify.py
sh build.sh
```

The verifier defaults to the independent bound 32 and brute-force bound 8. For a quick test:

```
python3 code/verify.py --max-n 12 --brute-n 7 \
  --output data/quick-check.json
```

The full run took about eleven seconds in the construction environment; this is a machine-dependent observation, not a complexity claim. The build script changes to its own directory before running pdfLaTeX, so it works even when invoked by an absolute path from another directory.

[write] These are the delivered names. In the collection the verifier is `code/02-nonholonomicity-verify.py`, its tests are `code/02-nonholonomicity-test_verify.py`, the build script is `code/02-nonholonomicity-build.sh`, and the records are `data/02-nonholonomicity-verification.json` and `data/02-nonholonomicity-unit-tests.txt`. Do not run them in place: the verifier's default output is `data/verification.json` under the report root, which is Part I's record; the renamed test file imports Part I's `code/verify.py` instead of the verifier it tests; and the build script looks for `article.tex` in `code/`. The report's README gives commands for a scratch copy with the delivered names. A rerun made that way when Part II was written passed, and its record differed from the shipped one only in the timing field.

25 Repository placement, dependency audit, and formalization

25.1 The inspected scope

The repository’s current main revision was resolved during this work as

8d936ee2357f9decf78c2ecc7d9baf3100a6787d.

The prior report is under

```
SetTheory/Cardinals/docs/reports/
generating-functions-and-asymptotics/
oeis-sequence-asymptotics/a003407-dyadic-scaling-rigidity/.
```

Its README explicitly distinguishes its unformalized research from the proved Ramsey-theoretic inputs [Part I, README; Section 11]. The present article is a continuation of its non-D-finiteness boundary, not a replacement of its profile, period, empirical-law, or gluing results. No repository files have been changed.

25.2 A claim-by-claim dependency map

Claim	Input and status
Deletion bound	Elementary injection, proved here.
Balanced splitting	Parity lower bound proved here; Sharma upper bound imported.
Dyadic nonconvergence	Published Ho result; reproduced using two published large counts.
Arithmetic regularity	Established G-function theorems, not proved from scratch here.
Finite-window rigidity	Complete Vandermonde proof in Section 19.
All arithmetic sections	Rigidity plus bounded-gap interpolation; complete proof.
Stable approximations	Complete logarithmic-error argument; arithmetic hypotheses explicit.
Positive observables	Complete proof using the existing two-sided shift estimate.
Continuous profile	Existing method, reproved as a short specialization; no novelty claim.
Small counts	Independently recomputed through 32, direct check through 8.
Lean/Rocq status	None of this article’s new results is formalized.

The prior report identifies formal versions of Sharma’s upper splitting bound and the two Davis et al. parity recurrences. Exact declaration names are recorded in the accompanying source audit. That report also identifies a native-computation trust boundary for the small count table [Part I, Section 11]. We did not run a Lean build or axiom audit. A conventional proof here does not acquire formal status by proximity to those declarations.

[write] The declaration names are in Part I’s Section 11; Subsection 11.4 says which of them Part II uses and records that the Lean `insertion_count_upper_bound_holds` is stronger than Lemma 17.1.

25.3 A feasible formalization boundary

A useful first formal target is the implication from a supplied leading layer (19.1) to the finite-window law and then to local-growth rigidity. It separates into the invertibility of a Vandermonde matrix, operator-norm inequalities, finite uniform $o(1)$ estimates, and elementary limits of powers and logarithms. This is a genuine theorem about supplied asymptotic data; it is not yet a formalization of the G-function theorem.

A second target is the purely discrete part: deletion, bounded-gap interpolation, binary descent, the rational power certificate, and the subset-state counting invariant. The main

result would then be a formally checked conditional theorem with the G-function asymptotic input explicitly named. A full unconditional formal proof would additionally require an appropriate development or formal import of arithmetic differential equations and their coefficient asymptotics. That missing infrastructure must not be concealed by naming an assumption as a theorem.

[write] Parts of the second target already exist in PROVEIT: the permutation definition, both splitting inequalities and a stronger insertion bound are Lean theorems (Subsection 11.4). Bounded-gap interpolation, binary descent, the power certificate, the subset-state invariant and everything in the first target are not formalized, and nothing of this plan has been executed.

26 Further research questions and topics

The following questions are proposed targets, not results asserted by this article. Their status is relative to the inspected sources and repository.

Question 26.1 (Analytic continuation and a natural boundary). Does the ordinary generating function F have a natural boundary on its circle of convergence? If not, determine a nontrivial continuation domain and describe the singularities encountered there.

The present argument removes D-finite models but does not supply the functional equation or singularity propagation needed for a natural-boundary proof. A productive first step would be to derive a generating-function identity from a refined state decomposition of the permutation class, rather than infer complex singularities from a real growth plot.

Question 26.2 (Differential algebraicity). Can F , or its entire exponential generating function E , satisfy a nonlinear polynomial differential equation over $\mathbb{C}(z)$?

The distinction is substantive: all linear equations have been excluded, while nonlinear equations can support much richer singular behavior. Any proposed equation should first be tested against exact coefficients and then against the two separated dyadic exponential scales.

Question 26.3 (Geometry and regularity of the growth profile). Determine the optimal modulus of continuity of P in (23.3), and decide where it is differentiable. Locate its global extrema and determine whether a maximum occurs at a dyadic phase.

The non-holonomicity proof needs only continuity and two unequal values. Finer profile information could instead guide sharp asymptotic algorithms, rigorous integration of the phase distribution, and extremal growth bounds. Existing period and modulus results in [Part I, Theorem 5.2, Proposition 3.3] should be kept as the baseline rather than rediscovered. [write] These are items 1 and 2 of Part I's Section 12.

Question 26.4 (Monotonicity and sharp adjacent ratios). Is $\theta(n+1) \geq \theta(n)$ always true? What is the smallest possible growth scale of the largest adjacent ratios?

Monotonicity is left open in the inspected enumerative sources. Our argument does not use it. The gap between a weak polynomial lower estimate and combinatorial positivity suggests investigating insertion positions and parity traces directly, rather than strengthening the analytic argument by an unsupported monotonicity assumption. [write] This is item 5 of Part I's Section 12 and Sharma's open problem (1), stated without proof in PROVEIT as `Sharma2012.open_problem_1`; the Lean `insertion_count_upper_bound_holds` bounds the adjacent ratio above by $\lfloor (n+3)/2 \rfloor$.

Question 26.5 (Signed observables and cancellation). Classify nonzero signed polynomials in finitely many shifted counts whose values remain positive or nonzero. Which of these evaluated sequences are P-recursive, if any?

The positivity proof of (21.1) does not extend to subtraction. An identically zero signed observable would of course be P-recursive, so a meaningful classification must separate exact identities from cancellation that changes the exponential scale. Even the asymptotic size of $\theta(n+1) - \theta(n)$ is relevant here.

Question 26.6 (An elementary substitute for the arithmetic theorem). For the particular splitting class (17.3), can non-P-recursive be proved directly from a hypothetical polynomial recurrence, without using the arithmetic regularity theorem for G-functions?

Such a proof could be more accessible to formalization. A plausible route is a rigorous classification of the pure-exponential portion of a rational coefficient difference system, with all possible factorial scales and cancellations controlled. A formal asymptotic ansatz alone would not suffice.

Question 26.7 (Certified large enumeration). Produce independently auditable certificates for $\theta(64)$ and $\theta(75)$, or replace them by a smaller structural separation that does not import a large count table.

The integer comparison in (18.4) is cheap; obtaining its inputs is the harder verification task. A certificate-producing subset-state algorithm with a small trusted checker would strengthen the computational boundary of both Ho’s theorem and the present application. [write] Still open in PROVEIT: its Lean count table `table_1_holds` stops at $\theta(1), \dots, \theta(20)$, behind a native trust boundary (Subsection 11.4). Item 7 of Part I’s Section 12 asks the same for the counts $100 \leq n \leq 200$ that Part I’s certificates use.

Question 26.8 (Effective recurrence obstructions). Given prescribed order, coefficient degree, and starting index, bound an interval of coefficients sufficient to disprove a recurrence for this sequence. Can such bounds be obtained from a quantitative finite-window argument rather than growing matrix searches?

The analytic contradiction is uniform over all hypothetical recurrences but does not supply a universal finite certificate size: its Vandermonde constants and asymptotic thresholds depend on that recurrence. Bridging this qualitative–effective gap is a separate problem.

Question 26.9 (Other hereditary enumerations with phase-dependent growth). Find other exponentially bounded, integer counting sequences with a polynomial insertion bound and a nonconstant logarithmic exponential profile, and apply Theorem 19.5 or Proposition 23.1.

The new criterion is not restricted to arithmetic-progressions. Its useful search signature is an exponential-scale phase oscillation together with local insertion control. Ordinary bounded multiplicative periodic factors are not enough: they leave n th-root growth unchanged.

27 Conclusion

The obstruction to holonomicity is not oscillation by itself, but the combination of two incompatible structures. Arithmetic P-recursive sequences of exponential size have a dominant finite-frequency power-logarithmic layer. A Vandermonde matrix forces that layer to be visible in every sufficiently late fixed-length window. The 3AP-free permutation counts, on the other

hand, cannot rise by an exponential factor in a bounded number of steps, yet they exhibit two different limiting exponential rates along dyadic scales.

This contradiction proves non-P-recursiveness of A003407 and every arithmetic section, non-D-finiteness of both generating functions, and stable versions for positive polynomial observables and subexponential multiplicative approximations. The established enumerative and G-function theorems remain visible dependencies. Natural boundaries, nonlinear differential equations, sharper profile structure, and full formalization remain distinct research targets.

A Minimal arithmetic certificate and reproducibility

The core exact-period certificate can be expressed as the following short fragment, where `theta` is the supplied dictionary of published counts. No logarithms or floating-point comparisons occur.

```
assert 2 * theta[128] * 25**128 > 57**128
assert 19**4 < 2 * 16**4
assert 2 * 128**2 < 182**2
for n in range(152, 183):
    assert 21 * theta[n] * 4**n < 9**n

assert (21*theta[192])**2 * 50**128 < \
    (2*theta[128])**3 * 49**128
assert (2*theta[129])**4 * 20**172 > \
    (21*theta[172])**3 * 21**172
```

The complete program validates the input structure, recomputes small counts, checks additional consistency and endpoint certificates, and writes a machine-readable record. To reproduce the executed checks and rebuild the article, run from the package directory:

```
python verify.py --max-n 32
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The existing figure is included in the package. Regeneration with `python plot_profile.py` additionally requires NumPy and Matplotlib; neither is needed for the exact arithmetic checks. The higher-precision printed decimals are supplementary to the strict rational enclosures. [write] These commands assume the delivered flat layout. In the collection the scripts are in `code/`, so run them on a copy as the report's README describes; running `code/verify.py` in place fails to find its data.

B A compact proof certificate at the mathematical level

[write] Appendices B and C belong to Part II; they are the manuscript's Appendices A and B.

The base theorem can be checked from the following chain without the profile construction or the two-sided shift estimate:

1. The sequence is positive and integral, and $\theta(n) \leq 21^{n-1}$.
2. Deleting the largest entry gives $\theta(n+j) \leq (n+j)^j \theta(n)$ for every fixed j .
3. Ho's two dyadic scales show that $\theta(n)^{1/n}$ does not converge.

4. A complex polynomial recurrence would descend to a rational one. Exponential size and integrality would make F a G-function.
5. Its nonzero leading Nilsson layer would imply, for some fixed s , $\max_{0 \leq j < s} \theta(n+j) = R^n \exp(o(n))$.
6. By item 2, that same maximum lies between $\theta(n)$ and $(n+s)^{s-1} \theta(n)$. Therefore $\theta(n)^{1/n} \rightarrow R$, contradicting item 3.

The only deep analytic step is the stated G-function input in item 5. The finite-window passage within that step is fully proved in Lemma 19.3. The original question is thereby reduced to a short contradiction, without silently replacing a limsup by a limit.

C Notation and index conventions

Symbol	Meaning
$\theta(n)$	3AP-free permutations of $[n]$; $\theta(0) = 1$.
a_n	$\log \theta(n)$, used only for $n \geq 1$.
τ_n	Exact logarithmic splitting toll, $n \geq 2$.
L, U, D	$\log 2$, $\log 21$, and $\log(21/2)$.
Δ_n	$a_{n+1} - a_n$; it is not assumed nonnegative.
ℓ_m	Limit of $a_{m2^t}/(m2^t)$ as $t \rightarrow \infty$.
R	Exponential scale of a hypothetical Nilsson layer, not a radius.
ζ_j	Distinct phases of modulus one in that layer.
s	Number of its nonzero dominant frequencies in Section 19; elsewhere an explicitly fixed shift bound.
q, r	Arithmetic-section step and residue parameter; r need not be $< q$.
P	Continuous 1-periodic logarithmic growth profile.
$C(S)$	Number of valid completions from subset state S .

Finite alterations of initial values do not affect P-recursiveness. All limits and logarithmic estimates concerning a zero or small initial index are understood on the eventually positive tail. [write] Section 15.3 lists where these symbols differ from Part I's.

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